

BCH212 Medical Biochemistry

Summative Assessment II — Simulation Exam

Lectures 10 – 21

Lecture	Topic	Real Exam	This Exam	[PP] Qs
Lec 10	DNA & RNA Synthesis	8 Qs	12 Qs	10 Qs
Lec 11	Protein Synthesis & Transport Protein	8 Qs	12 Qs	10 Qs
Lec 12	Liver Function	12 Qs	16 Qs	11 Qs
Lec 13	Thalassemia	6 Qs	10 Qs	8 Qs
Lec 14	Apoptosis	6 Qs	10 Qs	8 Qs
Lec 15	Human Genome & Gene Regulation	8 Qs	12 Qs	9 Qs
Lec 16	Blood & Heme Metabolism	4 Qs	8 Qs	7 Qs
Lec 17	Plasma Protein & Blood Clotting Factor	8 Qs	12 Qs	9 Qs
Lec 18	Nutrition	12 Qs	16 Qs	12 Qs
Lec 19	Genetic Engineering	8 Qs	12 Qs	9 Qs
Lec 20	Free Radical & Antioxidant	4 Qs	8 Qs	5 Qs
Lec 21	Clinical Correlation	10 Qs	14 Qs	10 Qs
TOTAL		94 Qs	142 Qs	108 Qs

[PP] = drawn from past papers (Med35–37 for Lec11, varied for others) — study these first! Total: 142 questions. Answer key with explanations is on the LAST PAGES.

LECTURE 10: DNA & RNA SYNTHESIS

Q1. [PP] DNA replication results in two daughter molecules. Which correctly describes their strand composition?

- A. Each molecule has 2 new strands
 - B. Each molecule has 1 new and 1 original strand
 - C. Each molecule has 2 original strands
 - D. Both molecules are completely new
 - E. None of the above
-

Q2. [PP] Although prokaryotic nucleotide addition is ~20× faster, eukaryotes replicate larger genomes efficiently because:

- A. Less genetic material to copy
 - B. Fewer polymerase types
 - C. Helicase unwinds more easily
 - D. Multiple origins of replication (thousands per chromosome)
 - E. Eukaryotes have no exons
-

Q3. [PP] Okazaki fragments are BEST defined as:

- A. Short RNA primers on the leading strand
 - B. Discontinuous DNA segments synthesised on the lagging strand
 - C. Long continuous DNA on the leading strand
 - D. Telomeric repeat sequences
 - E. Protein-DNA complexes at the replication fork
-

Q4. [PP] Which statement about telomeres is TRUE?

- A. Located at the centre of chromosomes (centromere)
 - B. Degraded by telomerase
 - C. Do not contain functional protein-coding genes
 - D. Lengthen with every round of replication
 - E. All of the above are false
-

Q5. [PP] Base excision repair (BER) specifically repairs:

- A. Thymine dimers caused by UV light
 - B. Mismatched bases from replication errors
 - C. Single damaged bases from alkylation or oxidation
 - D. Double-strand breaks
 - E. Large helix-distorting adducts
-

Q6. [PP] Which chemotherapy drug inhibits mammalian Topoisomerase II?

- A. Rifampicin
 - B. Amanitin
 - C. Chloramphenicol
 - D. Adriamycin (Doxorubicin)
 - E. Tetracycline
-

Q7. [PP] Xeroderma pigmentosum (XP) is caused by a defect in which DNA repair pathway?

- A. Base excision repair
 - B. Mismatch repair
 - C. Nucleotide excision repair (NER)
 - D. Homologous recombination
 - E. Non-homologous end joining
-

Q8. [PP] Which correctly distinguishes replication from transcription?

- A. Transcription requires a primer; replication does not
 - B. Replication requires a primer; transcription does not
 - C. Both use the same polymerase enzyme
 - D. Transcription uses dNTPs as substrate
 - E. Both replicate the entire DNA molecule
-

Q9. [PP] The coding strand DNA sequence for mRNA 5'-CCAAUUGGCGCGUCCCCGUACGCC-3' is:

- A. 5' CCAAUUGGCGCGUCCCCGUACGCC 3'
 - B. 5' GGUUAACCGCGCAAGGGCAUGCGG 3'
 - C. 5' CCAATTGGCGCGTTCCCGTACGCC 3'
 - D. 3' CCAATTGGCGCGAACCCGTACGCC 5'
 - E. 3' GGTAACCGCGCAAGGGCATGCGG 5'
-

Q10. [PP] An mRNA vaccine for SARS-CoV-2 works by which mechanism?

- A. Deleting the viral target gene
 - B. Being translated by host ribosomes to produce spike protein that stimulates immunity
 - C. Binding to host DNA to drive transcription
 - D. Regulating host DNA directly
 - E. Integrating into the host genome via reverse transcription
-

Q11. Which enzyme introduces negative supercoiling in bacterial DNA and is the target of quinolone antibiotics?

- A. DNA ligase
 - B. DNA helicase
 - C. DNA gyrase (type II topoisomerase)
 - D. DNA polymerase I
 - E. DNA primase
-

Q12. Insertion or deletion of 1–2 nucleotides in a coding sequence causes which type of mutation?

- A. Silent mutation
 - B. Missense mutation
 - C. Nonsense mutation
 - D. Frameshift mutation
 - E. Conservative substitution
-

LECTURE 11: PROTEIN SYNTHESIS & TRANSPORT PROTEIN

Q13. [PP] Which molecule guides the ribosome-mRNA-nascent protein complex to the endoplasmic reticulum during protein synthesis?

- A. RF1
 - B. RF2
 - C. Signal Recognition Particle (SRP)
 - D. EF-2
 - E. IF2
-

Q14. [PP] In a promoter mutation, what is the PRIMARY (most directly) altered event?

- A. mRNA production
 - B. tRNA production
 - C. RNA polymerase structure
 - D. DNA polymerase activity
 - E. Promoter site binding affinity
-

Q15. [PP] Which antibiotic blocks translation in BOTH prokaryotes AND eukaryotes?

- A. Chloramphenicol
 - B. Tetracycline
 - C. Puromycin
 - D. Streptomycin
 - E. Erythromycin
-

Q16. [PP] Which protein does NOT require a signal sequence for proper targeting?

- A. Insulin
 - B. Growth hormone
 - C. Cytochrome P450 (ER membrane)
 - D. GLUT4
 - E. Pyruvate kinase
-

Q17. [PP] Given mRNA: 5'-AUG UGG CAU CGU UAA-3'. How many amino acids does the translated peptide contain?

- A. 3
 - B. 4
 - C. 5
 - D. 6
 - E. 7
-

Q18. [PP] Tetracycline exerts its antibacterial effect by:

- A. Inhibiting peptidyl transferase in the 50S subunit
 - B. Blocking aminoacyl-tRNA binding to the 30S ribosomal A site
 - C. Causing misreading of mRNA
 - D. Inhibiting translocation by EF-G
 - E. Cleaving 16S rRNA
-

Q19. [PP] Chloramphenicol inhibits bacterial translation by blocking:

- A. Aminoacyl-tRNA synthetase
 - B. Peptidyl transferase activity in the 50S subunit
 - C. Translocation by EF-G
 - D. mRNA decoding at 30S
 - E. Ribosome assembly
-

Q20. [PP] Eukaryotic ribosome size is:

- A. 50S
 - B. 60S (large subunit only)
 - C. 70S
 - D. 80S
 - E. 100S
-

**Q21. [PP] From the following mRNA sequence, how many amino acids are found in Peptide X?
5'-GGGAUG CCU AAU GAG UGA CCU-3'**

- A. 2
 - B. 3
 - C. 4
 - D. 5
 - E. 6
-

Q22. [PP] Which elongation factor escorts aminoacyl-tRNA to the ribosome A site during translation elongation?

- A. IF2-GTP
 - B. EF-Ts-GDP
 - C. EF-Tu-GTP (EF1A in eukaryotes)
 - D. EF-G (eEF2)
 - E. RF2
-

Q23. In eukaryotic protein translation, the initiator Met-tRNA occupies which ribosomal site during initiation?

- A. E site (exit)
 - B. A site (aminoacyl)
 - C. P site (peptidyl)
 - D. Anywhere, then moves to P
 - E. Outside the ribosome until 60S joins
-

Q24. Post-translational modifications include all of the following EXCEPT:

- A. Glycosylation
 - B. Phosphorylation
 - C. Ubiquitination
 - D. Signal peptide cleavage
 - E. Codon degeneracy
-

LECTURE 12: LIVER FUNCTION

Q25. [PP] Which is the rate-limiting enzyme of bile acid synthesis?

- A. HMG-CoA reductase
 - B. Cholesterol 7 α -hydroxylase (CYP7A1)
 - C. Sterol 27-hydroxylase
 - D. UDP-glucuronyl transferase
 - E. Heme oxygenase
-

Q26. [PP] Which molecule conjugates bilirubin to make it water-soluble for excretion?

- A. Fatty acid
 - B. Acetyl-CoA
 - C. Glucuronic acid
 - D. Sulphate
 - E. Albumin
-

Q27. [PP] In thalassemia, the predominant form of elevated serum bilirubin is:

- A. Conjugated bilirubin
 - B. Unconjugated bilirubin
 - C. Bilirubin bound to apolipoprotein
 - D. Bilirubin bound to bile acids
 - E. Bilirubin bound to globulin
-

Q28. [PP] A 50-year-old with jaundice and bile duct gallstones most likely shows elevation of:

- A. Serum albumin
 - B. GGT and Alkaline phosphatase (ALP)
 - C. Alpha-fetoprotein
 - D. Glucuronic acid
 - E. ALT and AST
-

Q29. [PP] Disulfiram (Antabuse) treats alcoholism by inhibiting:

- A. Alcohol dehydrogenase (ADH)
 - B. Aldehyde dehydrogenase (ALDH)
 - C. CYP2E1 (MEOS)
 - D. Acetyl-CoA synthetase
 - E. Pyruvate dehydrogenase
-

Q30. [PP] Excess NADH generated by ethanol metabolism most directly causes:

- A. Increased gluconeogenesis
 - B. Activated TCA cycle
 - C. Increased fatty acid synthesis contributing to fatty liver
 - D. Decreased lactic acid
 - E. Increased fatty acid beta-oxidation
-

Q31. [PP] Which is the tumour marker for liver cancer (hepatocellular carcinoma)?

- A. Albumin
 - B. Prothrombin
 - C. Alpha-fetoprotein (AFP)
 - D. Haptoglobin
 - E. CRP
-

Q32. [PP] Which is a secondary bile acid (formed in the intestine by bacterial action)?

- A. Cholic acid
 - B. Chenodeoxycholic acid
 - C. Lithocholic acid
 - D. Glycocholic acid
 - E. Taurocholic acid
-

Q33. [PP] Phase II biotransformation in the liver is characterised by:

- A. Oxidation by CYP450
 - B. Reduction reactions
 - C. Conjugation reactions (glucuronidation, glutathione, sulphation)
 - D. Hydrolysis
 - E. Hydroxylation
-

Q34. [PP] Crigler-Najjar syndrome Type I results from complete deficiency of:

- A. Heme oxygenase
 - B. Biliverdin reductase
 - C. UDP-glucuronyl transferase (UGT1A1)
 - D. Cholesterol 7 α -hydroxylase
 - E. ALT
-

Q35. [PP] The enzyme responsible for the MEOS (Microsomal Ethanol Oxidising System) is:

- A. Alcohol dehydrogenase
 - B. Aldehyde dehydrogenase
 - C. CYP2E1
 - D. CYP3A4
 - E. Xanthine oxidase
-

Q36. Which test BEST reflects hepatic SYNTHETIC function?

- A. ALT
 - B. GGT
 - C. Alkaline phosphatase (ALP)
 - D. Serum albumin and prothrombin time (PT)
 - E. Total bilirubin
-

Q37. Heme oxygenase catalyses which reaction in bilirubin metabolism?

- A. Biliverdin $\rightarrow$ bilirubin
 - B. Haemoglobin $\rightarrow$ haem
 - C. Haem $\rightarrow$ biliverdin (opens porphyrin ring, rate-limiting step)
 - D. Bilirubin + glucuronic acid $\rightarrow$ conjugated bilirubin
 - E. Bilirubin $\rightarrow$ urobilinogen
-

Q38. In which type of jaundice is stool pale (clay-coloured) and urine dark, with urobilinogen absent from urine?

- A. Pre-hepatic (haemolytic)
 - B. Hepatic (hepatocellular)
 - C. Post-hepatic (obstructive/cholestatic)
 - D. Neonatal physiological jaundice
 - E. Gilbert's syndrome
-

Q39. Enterohepatic circulation of bile acids refers to:

- A. Portal blood flow from intestine to liver
 - B. Recycling of bile acids between liver, bile ducts, intestine, and back to liver via portal vein
 - C. Transport of glucose from intestine to liver
 - D. Lymphatic drainage of the liver
 - E. Enterocyte renewal
-

Q40. AST/ALT ratio >2 is most suggestive of:

- A. Acute viral hepatitis
 - B. Cholestasis
 - C. Alcoholic liver disease
 - D. Hepatocellular carcinoma
 - E. Haemolytic jaundice
-

LECTURE 13: THALASSEMIA

Q41. [PP] What is the genotype of Hb Bart's hydrops fetalis?

- A. $--/\alpha$ (3-gene deletion = HbH disease)
- B. $-\alpha/-\alpha$ (homozygous alpha-thal 2)
- C. $--/--$ (all 4 alpha genes absent)
- D. $\alpha\alpha/--$ (alpha-thal 1 trait)
- E. $\alpha^{CS}\alpha/--$

Q42. [PP] Which genotype results in Hb H disease?

- A. $--/--$ (all 4 deleted)
- B. $--/\alpha$ (3 of 4 alpha genes deleted)
- C. $-\alpha/-\alpha$ (2 deleted, one each chromosome)
- D. $\alpha\alpha/--$ (alpha-thal 1 trait)
- E. $\alpha\alpha/\alpha\alpha$ (normal)

Q43. [PP] Hydroxyurea is used in beta-thalassemia because it:

- A. Chelates excess iron from transfusions
- B. Pharmacologically reactivates gamma-globin gene expression → increased HbF
- C. Inhibits alpha chain synthesis
- D. Directly replaces deficient beta chains
- E. Suppresses erythropoiesis

Q44. [PP] Father has beta-thalassemia trait (carrier); mother has homozygous HbE. What is the probability their child has thalassemia disease?

- A. 0%
- B. 25%
- C. 50%
- D. 75%
- E. 100%

Q45. [PP] Hemoglobin Constant Spring (HbCS) is caused by which mutation?

- A. Deletion of both alpha genes on one chromosome
- B. Codon 26 GAG→AAG in the beta-globin gene
- C. Stop codon mutation at alpha-globin codon 142: UAA→CAA
- D. Frameshift in exon 2 of alpha-globin
- E. Splice-site mutation causing exon skipping

Q46. [PP] Which haemoglobin predominates in FETAL circulation?

- A. HbA (alpha β)
- B. HbF (alpha γ)
- C. HbA2 (alpha δ)
- D. Hb Gower 1 (zeta ϵ)
- E. Hb Portland (zeta γ)

Q47. [PP] Which haemoglobin is found in the EMBRYO (earliest development, yolk sac stage)?

- A. alpha β (HbA)
- B. alpha γ (HbF)
- C. alpha δ (HbA2)
- D. alpha ζ
- E. zeta ϵ (Hb Gower 1)

Q48. [PP] Repeated blood transfusions in severe thalassemia most commonly leads to:

- A. Hyperkalaemia
 - B. Iron overload (haemosiderosis) damaging heart, liver, and endocrine glands
 - C. Folate deficiency
 - D. Elevated HbF
 - E. Thrombocytopenia
-

Q49. In beta-zero thalassemia/HbE (most common severe thalassemia in Thailand), the HbE mutation at codon 26 (GAG→AAG) also creates:

- A. A new start codon
 - B. A cryptic splice site in exon 1, causing reduced normal beta-globin mRNA
 - C. A premature stop codon
 - D. A frameshift mutation
 - E. A promoter deletion
-

Q50. Alpha-thalassemia trait (2 functional alpha genes remain) can result from which genotype(s)?

- A. $--/\alpha\alpha$ (cis, alpha-thal 1 heterozygote) only
 - B. $-\alpha/-\alpha$ (trans, alpha-thal 2 homozygote) only
 - C. Either $--/\alpha\alpha$ or $-\alpha/-\alpha$
 - D. $--/--$ only
 - E. $\alpha^{\wedge}CS \alpha/\alpha\alpha$ only
-

LECTURE 14: APOPTOSIS

Q51. [PP] Which is a characteristic of APOPTOSIS but NOT necrosis?

- A. Cell swelling
 - B. Chromatin condensation and nuclear fragmentation
 - C. Membrane rupture releasing cell contents
 - D. Significant inflammation
 - E. Random DNA degradation
-

Q52. [PP] Which component is NOT part of the apoptosome?

- A. Cytochrome C
 - B. APAF-1
 - C. dATP
 - D. Pro-caspase 9
 - E. Pro-caspase 10
-

Q53. [PP] Which MAPK signalling kinase INDUCES apoptosis?

- A. ERK1/2 (promotes survival/proliferation)
 - B. p38 MAPK (stress-activated, pro-apoptotic)
 - C. AKT/PKB (survival kinase)
 - D. PDK1
 - E. mTOR (growth)
-

Q54. [PP] Which member of the Bcl-2 family is PRO-apoptotic?

- A. Bcl-2
 - B. BAX
 - C. Mcl-1
 - D. Bcl-w
 - E. Bcl-xL
-

Q55. [PP] Hoechst 33342 staining is used in apoptosis research to assess:

- A. Mitochondrial membrane potential
 - B. Phosphatidylserine externalisation
 - C. Nuclear morphology (chromatin condensation and fragmentation)
 - D. DNA content by flow cytometry
 - E. Protein expression
-

Q56. [PP] In flow cytometric cell cycle analysis, apoptotic cells appear as which population?

- A. G0/G1 peak
 - B. Sub-G1 peak (left of G1)
 - C. S phase fraction
 - D. G2/M peak
 - E. Hyperdiploid peak
-

Q57. [PP] The KEY initiator caspase of the EXTRINSIC (death receptor) apoptosis pathway is:

- A. Caspase 3
 - B. Caspase 6
 - C. Caspase 7
 - D. Caspase 8
 - E. Caspase 9
-

Q58. [PP] Which technique detects apoptosis by identifying phosphatidylserine externalisation?

- A. Western blot for PARP cleavage
 - B. Hoechst 33342 staining
 - C. Flow cytometry with fluorochrome-labelled Annexin V
 - D. Cell cycle analysis with propidium iodide
 - E. Electron microscopy
-

Q59. PARP cleavage by caspases during apoptosis results in:

- A. Activated DNA repair → cell survival
 - B. Inactivated DNA repair → promotes commitment to cell death
 - C. Release of cytochrome C from mitochondria
 - D. Activation of initiator caspases
 - E. Formation of apoptotic bodies
-

Q60. Anti-apoptotic Bcl-2 family proteins (e.g., Bcl-2, Bcl-xL) prevent apoptosis by:

- A. Directly inhibiting caspase 3 and 7
 - B. Blocking BAX/BAK pore formation in the outer mitochondrial membrane
 - C. Blocking death receptor signalling at the cell surface
 - D. Degrading APAF-1
 - E. Exporting cytochrome C from the cell
-

LECTURE 15: HUMAN GENOME & GENE REGULATION

Q61. [PP] Fragile X syndrome CGG repeat expansion in FMR1 primarily disrupts which process?

- A. Splicing
 - B. Translation
 - C. Replication
 - D. Transcription (silenced by hypermethylation)
 - E. Post-translational modification
-

Q62. [PP] A 25-year-old female with infertility, no menarche, webbed neck, and broad chest. Most likely karyotype?

- A. 46, XO
 - B. 45, XO (Turner syndrome)
 - C. 47, XXX
 - D. 47, XXY
 - E. 46, XX, t(14;21)
-

Q63. [PP] The Mediator complex in eukaryotic transcription functions as:

- A. An RNA polymerase subunit
 - B. A bridge connecting activators at enhancers to transcription factors at the promoter
 - C. A sigma factor equivalent
 - D. A repressor protein
 - E. A spliceosome component
-

Q64. [PP] Estrogen receptor (ER) is classified as which receptor type?

- A. Nuclear receptor (ligand-activated transcription factor)
 - B. Cell membrane receptor
 - C. Receptor tyrosine kinase
 - D. Ion channel receptor
 - E. G-protein coupled receptor
-

Q65. [PP] A promoter mutation most DIRECTLY alters which event?

- A. DNA polymerase activity
 - B. RNA polymerase binding efficiency
 - C. mRNA production rate
 - D. Protein production
 - E. Promoter site binding affinity for transcription factors/RNA polymerase
-

Q66. [PP] For MAXIMAL lac operon expression in E. coli, which conditions are required?

- A. High cAMP and no lactose
 - B. Lactose present AND glucose absent (CAP site occupied)
 - C. High glucose and high lactose
 - D. Attenuation stem-loop formation
 - E. Repressor bound to operator
-

Q67. [PP] Klinefelter syndrome (47,XXY) is caused by:

- A. Deletion of X chromosome
 - B. X chromosome inversion
 - C. Non-disjunction during meiosis
 - D. Trinucleotide repeat expansion
 - E. Robertsonian translocation
-

Q68. [PP] A riboswitch controls gene expression at the TRANSCRIPTIONAL level when metabolite binding causes:

- A. Ribosome stalling on the mRNA
 - B. Formation of a terminator stem-loop + poly-U track → premature termination
 - C. mRNA degradation by RNase E
 - D. Increased mRNA stability
 - E. Sigma factor recruitment
-

Q69. [PP] AGO2-mediated mRNA cleavage in the siRNA pathway requires:

- A. Perfect complementarity between siRNA and target mRNA
 - B. Imperfect complementarity (>30% mismatches)
 - C. DNA methylation of the target gene
 - D. Histone deacetylation
 - E. Nuclear localisation of AGO2
-

Q70. DNA methylation at CpG promoter regions typically results in:

- A. Gene activation
 - B. Histone acetylation → open chromatin
 - C. Gene silencing via heterochromatin formation
 - D. Increased RNA polymerase binding
 - E. Enhanced enhancer activity
-

Q71. Which feature makes VNTR/STR markers useful for DNA fingerprinting?

- A. All humans have identical VNTR sequences
 - B. VNTRs code for essential proteins
 - C. The number of repeats varies between individuals, creating unique patterns
 - D. VNTRs are only on sex chromosomes
 - E. VNTRs cannot be amplified by PCR
-

Q72. Chromosomal notation for Down syndrome (trisomy 21) in a female is:

- A. 45,XX
 - B. 46,XX
 - C. 47,XX,+21
 - D. 47,XXY
 - E. 46,XX,t(14;21)
-

LECTURE 16: BLOOD & HEME METABOLISM

Q73. [PP] The rate-limiting enzyme of haem synthesis is:

- A. ALA dehydratase (ALAD)
 - B. Delta-aminolevulinic acid synthase (ALAS)
 - C. Porphobilinogen deaminase
 - D. Ferrochelatase
 - E. Uroporphyrinogen decarboxylase
-

Q74. [PP] Which TCA cycle compound is the direct precursor for haem synthesis?

- A. Oxaloacetate
 - B. Citrate
 - C. Succinyl-CoA
 - D. Fumarate
 - E. Malate
-

Q75. [PP] Lead (Pb) poisoning inhibits which TWO enzymes in haem synthesis?

- A. ALAS and ferrochelatase
 - B. ALA dehydratase (ALAD) and ferrochelatase
 - C. Porphobilinogen deaminase and UROD
 - D. ALAS and uroporphyrinogen synthase
 - E. Ferrochelatase and porphobilinogen deaminase
-

Q76. [PP] Porphyria Cutanea Tarda (PCT), the most common porphyria, is caused by deficiency of:

- A. ALAS
 - B. ALAD
 - C. Porphobilinogen deaminase
 - D. Uroporphyrinogen decarboxylase (UROD)
 - E. Ferrochelatase
-

Q77. [PP] Acute Intermittent Porphyria (AIP) is characterised by all of the following EXCEPT:

- A. Acute abdominal pain
 - B. Neuropsychiatric symptoms
 - C. Port-wine coloured urine
 - D. Photosensitivity and skin blistering
 - E. Precipitated by barbiturates
-

Q78. [PP] Which enzyme in haem synthesis requires Vitamin B6 (pyridoxal phosphate) as cofactor?

- A. ALA synthase (ALAS)
 - B. ALA dehydratase
 - C. Porphobilinogen deaminase
 - D. Uroporphyrinogen decarboxylase
 - E. Ferrochelatase
-

Q79. [PP] Mature red blood cells CANNOT synthesise haem because they:

- A. Lack all haem synthesis enzymes
 - B. Have no mitochondria (ALAS is mitochondrial — first step cannot occur)
 - C. Lack succinyl-CoA
 - D. Have no DNA
 - E. Both B and C
-

Q80. Which amino acid, together with succinyl-CoA, is the starting material for haem synthesis?

- A. Alanine
 - B. Serine
 - C. Glycine
 - D. Tyrosine
 - E. Cysteine
-

LECTURE 17: PLASMA PROTEIN & BLOOD CLOTTING FACTOR

Q81. [PP] The MAJOR plasma protein and its primary function are:

- A. Prealbumin — thyroxine transport
 - B. Albumin — maintains plasma oncotic (colloid osmotic) pressure
 - C. Globulin — primary immune function
 - D. Fibrinogen — blood clotting
 - E. AFP — fetal development
-

Q82. [PP] The FIRST step in haemostasis after blood vessel injury is:

- A. Fibrin clot formation
 - B. Platelet aggregation
 - C. Blood vessel constriction (vasoconstriction)
 - D. Dense granule secretion of ADP
 - E. Fibrinolysis
-

Q83. [PP] Which enzyme digests fibrin during fibrinolysis?

- A. Fibrinogen
 - B. Plasmin
 - C. Dicoumarol
 - D. Thrombin
 - E. Protein C
-

Q84. [PP] Alpha-1 antitrypsin (AAT) deficiency primarily causes:

- A. Multiple myeloma
 - B. Von Willebrand disease
 - C. Emphysema (panacinar) and liver cirrhosis
 - D. Haemophilia A
 - E. DIC
-

Q85. [PP] Multiple myeloma is associated with elevated levels of which plasma protein fraction?

- A. Albumin (elevated)
 - B. Alpha-1 globulin
 - C. Alpha-2 globulin
 - D. Gamma globulin (monoclonal immunoglobulins = M-protein)
 - E. Prealbumin
-

Q86. [PP] Aspirin prevents arterial thrombosis by:

- A. Promoting fibrinolysis
 - B. Irreversibly inhibiting COX → blocks thromboxane A₂ (TXA₂) synthesis in platelets
 - C. Blocking vitamin K recycling
 - D. Inhibiting thrombin formation
 - E. Activating antithrombin III
-

Q87. [PP] Vitamin K deficiency most directly leads to:

- A. Increased plasma prothrombin
 - B. Prolonged prothrombin time (PT) due to deficiency of functional factors II, VII, IX, X
 - C. Thrombocytopenia
 - D. Elevated fibrinogen
 - E. Decreased antithrombin III
-

Q88. [PP] Haemophilia A is caused by deficiency of:

- A. Factor V
 - B. Factor VII
 - C. Factor VIII
 - D. Factor IX
 - E. Factor X
-

Q89. [PP] Von Willebrand factor (vWF) plays what primary role in haemostasis?

- A. Activates the intrinsic coagulation cascade
 - B. Bridges platelet GP Ib/IX receptor to exposed subendothelial collagen (platelet adhesion)
 - C. Converts fibrinogen to fibrin
 - D. Activates protein C
 - E. Inhibits thrombin
-

Q90. Haptoglobin's main function in plasma is to:

- A. Transport Fe^{3+} in blood
 - B. Bind free haemoglobin from haemolysed RBCs, preventing kidney damage and iron loss
 - C. Transport copper
 - D. Inhibit elastase
 - E. Activate complement
-

Q91. Albumin does NOT transport which of the following?

- A. Fatty acids
 - B. Unconjugated bilirubin
 - C. Iron (Fe^{3+})
 - D. Drugs (warfarin, NSAIDs)
 - E. Calcium (40% bound fraction)
-

Q92. t-PA (tissue plasminogen activator) causes:

- A. Fibrinogen → fibrin (clot formation)
 - B. Platelet aggregation via GPIIb/IIIa
 - C. Plasminogen → plasmin (fibrinolysis)
 - D. Activation of Factor X
 - E. Inhibition of thrombin
-

LECTURE 18: NUTRITION

Q93. [PP] Pellagra (dermatitis, diarrhoea, dementia) is caused by deficiency of:

- A. Thiamine (B1)
 - B. Riboflavin (B2)
 - C. Niacin (B3)
 - D. Pyridoxine (B6)
 - E. Cobalamin (B12)
-

Q94. [PP] Prolonged consumption of raw egg whites causes deficiency of:

- A. Thiamine (B1)
 - B. Riboflavin (B2)
 - C. Niacin (B3)
 - D. Biotin (B7)
 - E. Folate (B9)
-

Q95. [PP] Macrocytic anaemia + elevated homocysteine + NORMAL methylmalonic acid indicates:

- A. Vitamin B12 deficiency
 - B. Folate (B9) deficiency
 - C. Pyridoxine (B6) deficiency
 - D. Vitamin C deficiency
 - E. Iron deficiency anaemia
-

Q96. [PP] Vitamin D deficiency results in:

- A. Hypercalcaemia and hyperphosphataemia
 - B. Hypocalcaemia and hypophosphataemia → rickets (children) / osteomalacia (adults)
 - C. Night blindness
 - D. Megaloblastic anaemia
 - E. Scurvy
-

Q97. [PP] Vitamin C is required as cofactor for which enzyme in collagen synthesis?

- A. Carboxylase
 - B. Prolyl/Lysyl Hydroxylase
 - C. Decarboxylase
 - D. Hydratase
 - E. Phosphokinase
-

Q98. [PP] Cretinism in newborns results from severe maternal deficiency of:

- A. Iron
 - B. Iodine
 - C. Folate
 - D. Vitamin B12
 - E. Zinc
-

Q99. [PP] Neural tube defects (spina bifida, anencephaly) result from deficiency of:

- A. Folate (B9)
 - B. Vitamin B12
 - C. Vitamin D
 - D. Vitamin A
 - E. Biotin
-

Q100. [PP] Isoniazid (INH) treatment for TB most directly causes secondary deficiency of:

- A. Thiamine (B1)
 - B. Niacin (B3) directly
 - C. Pyridoxine (B6)
 - D. Cobalamin (B12)
 - E. Vitamin C
-

Q101. [PP] The ACTIVE form of Vitamin D produced in the KIDNEY is:

- A. Cholecalciferol (D3)
 - B. Ergocalciferol (D2)
 - C. 25-hydroxycalciferol (calcidiol, storage form)
 - D. 1,25-dihydroxycalciferol (calcitriol, active form)
 - E. 24,25-dihydroxycalciferol
-

Q102. [PP] Kwashiorkor features oedema but NOT Marasmus because kwashiorkor involves:

- A. More severe total calorie restriction
 - B. Protein deficiency → low albumin → low oncotic pressure → oedema
 - C. Higher rates of infection
 - D. Affecting younger infants only
 - E. More severe liver damage
-

Q103. [PP] Prolonged PPI (proton pump inhibitor) use most commonly leads to deficiency of:

- A. Thiamine (B1)
 - B. Folate (B9)
 - C. Vitamin B12
 - D. Vitamin D
 - E. Vitamin K
-

Q104. [PP] Which statement about fat-soluble vitamins (A, D, E, K) is TRUE?

- A. They mainly act as coenzymes (like B vitamins)
 - B. They are stored in body fat/liver and can accumulate to toxic levels
 - C. They are easily excreted in urine
 - D. They require frequent daily dietary intake
 - E. Excessive intake is always safe
-

Q105. Vitamin K is required for gamma-carboxylation of glutamate residues in which clotting factors?

- A. Serine residues in Factor XIII
 - B. Glutamate residues in Factors II, VII, IX, X (and Proteins C, S)
 - C. Lysine residues in Factor VIII
 - D. Cysteine residues in Factor V
 - E. Threonine residues in fibrinogen
-

Q106. Vitamin C acts as an antioxidant primarily by:

- A. Inhibiting free radical chain reactions AND regenerating active Vitamin E (tocopherol) from tocopheroxyl radical
 - B. Directly decomposing hydrogen peroxide
 - C. Chelating iron to prevent Fenton reaction
 - D. Inhibiting xanthine oxidase
 - E. Activating superoxide dismutase
-

Q107. A 4-year-old with bleeding gums, perifollicular haemorrhages, poor wound healing, and bone pain. Which vitamin is deficient?

- A. Vitamin B2
 - B. Vitamin B9 (Folate)
 - C. Vitamin C
 - D. Vitamin D
 - E. Vitamin K
-

Q108. Which vitamin is directly involved in converting tryptophan to niacin?

- A. Thiamine (B1)
 - B. Riboflavin (B2)
 - C. Pyridoxine (B6)
 - D. Cobalamin (B12)
 - E. Vitamin D
-

LECTURE 19: GENETIC ENGINEERING

Q109. [PP] Which technique BEST quantifies SARS-CoV-2 viral load?

- A. ATK (antigen rapid test)
 - B. Conventional PCR
 - C. RT-PCR (qualitative)
 - D. RT-qPCR (reverse transcription quantitative PCR)
 - E. ELISA (antibody detection)
-

Q110. [PP] In forensic DNA analysis, which technique creates a DNA 'fingerprint' using VNTR probes?

- A. Northern blot
 - B. Southern blot
 - C. Western blot
 - D. RT-PCR
 - E. DNA microarray
-

Q111. [PP] High-throughput gene expression profiling across thousands of genes simultaneously uses:

- A. FISH
 - B. RT-PCR (single gene)
 - C. PCR-RFLP
 - D. DNA microarray
 - E. Sanger sequencing
-

Q112. [PP] Which method introduces foreign DNA into cells using an electrical field to create temporary membrane pores?

- A. Electrophoresis
 - B. Electroporation
 - C. Heat shock transformation
 - D. Lipofection
 - E. Viral transduction
-

Q113. [PP] Why must RT-PCR be used to clone a eukaryotic gene for expression in bacteria?

- A. Bacteria grow faster than eukaryotes
 - B. Bacteria lack spliceosomes to remove introns; RT-PCR generates intron-free cDNA from mRNA
 - C. PCR cannot amplify eukaryotic DNA
 - D. Bacteria require RNA as template
 - E. Eukaryotic DNA is too large for PCR
-

Q114. [PP] DNA ligase joins two DNA fragments by forming:

- A. Hydrogen bond
 - B. Phosphodiester bond between 3'-OH and 5'-phosphate
 - C. Peptide bond
 - D. Glycosidic bond
 - E. Van der Waals interaction
-

Q115. [PP] The FASTEST method to screen bacterial colonies for correct gene insertion is:

- A. Replica plating
 - B. Colony PCR
 - C. RT-PCR
 - D. Southern blotting
 - E. Sanger sequencing after plasmid extraction
-

Q116. [PP] Western blotting detects specific proteins using:

- A. Fluorescent DNA probes
 - B. VNTR probes
 - C. Antibody specific to the target protein
 - D. Complementary RNA probes
 - E. Restriction enzymes
-

Q117. [PP] Huntington's disease is caused by which mutation in the HTT gene?

- A. Missense point mutation (single amino acid change)
 - B. Large exon deletion
 - C. Excessive CAG trinucleotide repeat expansion (>36 repeats)
 - D. Promoter hypermethylation silencing HTT
 - E. Nonsense mutation causing truncated protein
-

Q118. EcoRI cuts 5'-G|AATTC-3'. What type of DNA ends are produced?

- A. Blunt ends
 - B. 5' overhangs (5' sticky ends)
 - C. 3' overhangs (3' sticky ends)
 - D. Nicks in one strand only
 - E. Circular ends
-

Q119. Which PCR primer design rule is INCORRECT?

- A. Primer length ~20 bases
 - B. GC content 45–55%
 - C. The 3' end should be G or C (GC clamp)
 - D. Primers should self-complement to form hairpins for efficient binding
 - E. Forward and reverse primers should have similar melting temperatures (T_m)
-

Q120. In blue-white screening, white bacterial colonies indicate:

- A. No plasmid was taken up
 - B. Successful gene insertion into the lacZ site (lacZ disrupted → no beta-galactosidase → no blue colour)
 - C. The gene was inserted in the wrong orientation
 - D. Antibiotic-resistant but no insert
 - E. Contamination
-

LECTURE 20: FREE RADICAL & ANTIOXIDANT

Q121. [PP] Which enzyme provides the FIRST LINE of enzymatic defence against superoxide ($\text{O}_2^{\bullet-}$)?

- A. Catalase
- B. Glutathione peroxidase (GPx)
- C. Superoxide dismutase (SOD)
- D. Glutathione reductase
- E. Peroxiredoxin

Q122. [PP] Which mineral is incorporated into glutathione peroxidase (GPx)?

- A. Iron (Fe)
- B. Manganese (Mn)
- C. Selenium (Se)
- D. Copper (Cu)
- E. Zinc (Zn)

Q123. [PP] Which vitamin regenerates oxidised Vitamin E (tocopheroxyl radical) back to its active form?

- A. Vitamin A
- B. Vitamin C (Ascorbic acid)
- C. Vitamin D
- D. Vitamin K
- E. Vitamin B12

Q124. [PP] A characteristic property of free radicals is all of the following EXCEPT:

- A. Extreme reactivity
- B. Small and diffusible
- C. Very short lifespan
- D. Long lifespan (hours to days)
- E. Generate new radicals by chain reaction

Q125. [PP] Which part of amino acids is MOST susceptible to attack by free radicals?

- A. R-group (general side chain)
- B. OH group
- C. Amino group ($-\text{NH}_2$)
- D. Carboxylic group ($-\text{COOH}$)
- E. Sulfhydryl group ($-\text{SH}$) of cysteine

Q126. Cell MEMBRANE antioxidants include:

- A. Vitamin C and uric acid (plasma)
- B. SOD and catalase (intracellular)
- C. Alpha-tocopherol (Vitamin E) and Coenzyme Q10
- D. Transferrin and ceruloplasmin (plasma)
- E. Glutathione peroxidase and glutathione (intracellular)

Q127. Oxidative stress is defined as:

- A. The presence of any free radicals in the body
- B. Complete absence of antioxidants
- C. Imbalance between ROS/free radical production and antioxidant defence (excess ROS)
- D. DNA damage from free radicals only
- E. Mitochondrial dysfunction only

Q128. Superoxide anion ($O_2^{\bullet-}$) is PRIMARILY produced by which cellular compartment?

- A. Nucleus
 - B. Mitochondria (electron transport chain leakage at Complex I and III)
 - C. Endoplasmic reticulum
 - D. Golgi apparatus
 - E. Lysosomes
-

LECTURE 21: CLINICAL CORRELATION

Q129. [PP] Which oil has the HIGHEST proportion of saturated fatty acids?

- A. Corn oil
 - B. Coconut oil
 - C. Olive oil
 - D. Canola oil
 - E. Soybean oil
-

Q130. [PP] The antidote for paracetamol (acetaminophen) overdose is:

- A. EDTA (chelation)
 - B. Deferoxamine (DFO)
 - C. Naloxone
 - D. Fomepizole
 - E. N-acetylcysteine (NAC)
-

Q131. [PP] After phototherapy for neonatal jaundice, which compound is secreted in BOTH bile and urine?

- A. Urobilinogen
 - B. Stercobilinogen
 - C. Unconjugated bilirubin
 - D. Lumirubin
 - E. Urobilin
-

Q132. [PP] Band keratopathy (calcium deposits in cornea) is associated with which metabolic condition?

- A. Obesity
 - B. Type 1 Diabetes
 - C. Hypocalcaemia
 - D. Hypercalcaemia
 - E. Hypothyroidism
-

Q133. [PP] Which endocrine-disrupting chemical (EDC) is commonly found in polycarbonate plastics?

- A. Dioxins
 - B. Lead
 - C. Bisphenol A (BPA)
 - D. Mercury
 - E. Arsenic
-

Q134. [PP] Which form of Vitamin D has the LONGEST half-life in blood?

- A. 1,25-dihydroxycalciferol (calcitriol, active form)
 - B. 25-hydroxycalciferol (calcidiol, storage form)
 - C. Vitamin D2 (ergocalciferol)
 - D. Vitamin D3 (cholecalciferol)
 - E. 7-dehydrocholesterol (precursor)
-

Q135. [PP] Neural tube defects (spina bifida) are caused by deficiency of which nutrient in early pregnancy?

- A. Iodine
 - B. Iron
 - C. Folate (Vitamin B9)
 - D. Vitamin B12
 - E. Zinc
-

Q136. [PP] Bitot's spots on the conjunctiva represent an early sign of deficiency of:

- A. Vitamin D
 - B. Vitamin A
 - C. Vitamin C
 - D. Vitamin K
 - E. Vitamin B12
-

Q137. [PP] In phenylketonuria (PKU), which amino acid is DECREASED in plasma?

- A. Alanine
 - B. Cysteine
 - C. Tyrosine
 - D. Glutamine
 - E. Phenylalanine (actually increased)
-

Q138. [PP] Standard OGTT (Oral Glucose Tolerance Test) uses what glucose dose?

- A. 25 g
 - B. 50 g
 - C. 75 g
 - D. 100 g
 - E. 150 g
-

Q139. A chronic alcoholic presents with oedema, dyspnoea, cardiomegaly, and metabolic acidosis. The most likely vitamin deficiency is:

- A. Niacin (B3)
 - B. Thiamine (B1)
 - C. Riboflavin (B2)
 - D. Vitamin B12
 - E. Vitamin C
-

Q140. Kayser-Fleischer rings + liver disease + neuropsychiatric symptoms. Diagnosis and deficient/elevated substance?

- A. Haemochromatosis — serum iron elevated
 - B. Wilson's disease — serum ceruloplasmin markedly low
 - C. Alpha-1 antitrypsin deficiency — AAT low
 - D. Porphyria cutanea tarda — UROD low
 - E. Gilbert's syndrome — UGT1A1 mildly reduced
-

Q141. Acanthosis nigricans (dark velvety skin at neck and axillae) in an obese teenager is most associated with:

- A. Vitamin A toxicity
 - B. Insulin resistance and hyperinsulinaemia activating IGF-1 receptors on keratinocytes
 - C. Addison's disease (ACTH-related hyperpigmentation)
 - D. Primary hypothyroidism
 - E. Cushing's syndrome
-

Q142. A patient with gallstones has elevated ALP and GGT but NORMAL ALT and AST. This pattern indicates:

- A. Acute viral hepatitis
 - B. Alcoholic hepatitis
 - C. Cholestasis (biliary obstruction)
 - D. Hepatocellular carcinoma
 - E. Haemolytic anaemia
-

ANSWER KEY & EXPLANATIONS

All answers with detailed explanations, organised by lecture. [PP] = past paper question.

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	B	2	D	3	B	4	C	5	C
6	D	7	C	8	B	9	C	10	B
11	C	12	D	13	C	14	E	15	C
16	E	17	B	18	B	19	B	20	D
21	B	22	C	23	C	24	E	25	B
26	C	27	B	28	B	29	B	30	C
31	C	32	C	33	C	34	C	35	C
36	D	37	C	38	C	39	B	40	C
41	C	42	B	43	B	44	C	45	C
46	B	47	E	48	B	49	B	50	C
51	B	52	E	53	B	54	B	55	C
56	B	57	D	58	C	59	B	60	B
61	D	62	B	63	B	64	A	65	E
66	B	67	C	68	B	69	A	70	C
71	C	72	C	73	B	74	C	75	B
76	D	77	D	78	A	79	B	80	C
81	B	82	C	83	B	84	C	85	D
86	B	87	B	88	C	89	B	90	B
91	C	92	C	93	C	94	D	95	B
96	B	97	B	98	B	99	A	100	C
101	D	102	B	103	C	104	B	105	B
106	A	107	C	108	C	109	D	110	B
111	D	112	B	113	B	114	B	115	B
116	C	117	C	118	B	119	D	120	B
121	C	122	C	123	B	124	D	125	E
126	C	127	C	128	B	129	B	130	E
131	D	132	D	133	C	134	B	135	C
136	B	137	C	138	C	139	B	140	B
141	B	142	C						

Lec 10: DNA & RNA Synthesis

Q1. [PP] DNA replication results in two daughter molecules. Which correctly describes their strand composition?

Answer: B

Semi-conservative replication: each daughter molecule retains one parental strand and gains one new strand — confirmed by the Meselson-Stahl experiment (1958). [PP: SA2 Med34 Q4]

Q2. [PP] Although prokaryotic nucleotide addition is ~20× faster, eukaryotes replicate larger genomes efficiently because:

Answer: D

Eukaryotes fire thousands of origins simultaneously across all chromosomes, compensating for slower per-site polymerisation. Prokaryotes use a single origin per chromosome. [PP: SA2 Med34 Q3]

Q3. [PP] Okazaki fragments are BEST defined as:

Answer: B

Because DNA polymerase adds only 5'→3', the lagging strand is copied discontinuously in short Okazaki fragments, each primed by a short RNA, then joined by DNA ligase. [PP: SA2 Med34 Q5]

Q4. [PP] Which statement about telomeres is TRUE?

Answer: C

Telomeres are repetitive non-coding sequences (TTAGGG ×~2500) at chromosome ends. Telomerase ELONGATES them (not degrades). Without telomerase, telomeres shorten with each division. [PP: SA2 Med34 Q1]

Q5. [PP] Base excision repair (BER) specifically repairs:

Answer: C

BER removes single damaged bases (alkylated, oxidised, deaminated) using a glycosylase, then replaces the nucleotide. NER removes bulky UV-induced lesions. MMR corrects replication mismatches. [PP: Key Simulation Q1]

Q6. [PP] Which chemotherapy drug inhibits mammalian Topoisomerase II?

Answer: D

Adriamycin inhibits topoisomerase II, preventing DNA strand re-ligation → lethal double-strand breaks in rapidly dividing cancer cells. Rifampicin targets bacterial RNA polymerase; Amanitin targets eukaryotic RNA-pol II. [PP: Key Simulation Q3]

Q7. [PP] Xeroderma pigmentosum (XP) is caused by a defect in which DNA repair pathway?

Answer: C

XP (autosomal recessive): NER normally removes thymine dimers from UV damage. Without NER, UV-induced lesions accumulate → multiple skin cancers at a young age. [PP: FA2 Sec11 Q3]

Q8. [PP] Which correctly distinguishes replication from transcription?

Answer: B

DNA polymerase cannot start de novo — it requires a short RNA primer (made by primase). RNA polymerase initiates transcription without a primer. Transcription uses NTPs (ribonucleotides); replication uses dNTPs. [PP: Key Simulation Q4]

Q9. [PP] The coding strand DNA sequence for mRNA 5'-CCAAUUGGCGCGUUCCCGUACGCC-3' is:

Answer: C

The coding (sense) strand mirrors the mRNA but with T instead of U. Replace all U→T: 5'-CCAATTGGCGCGTCCCGTACGCC-3'. [PP: SA2 Med34 Q2]

Q10. [PP] An mRNA vaccine for SARS-CoV-2 works by which mechanism?

Answer: B

Lipid nanoparticle-encased mRNA is translated by host ribosomes → spike protein → immune response. The mRNA never enters the nucleus or alters DNA and is eventually degraded. [PP: SA2 MED35 Q8]

Q11. Which enzyme introduces negative supercoiling in bacterial DNA and is the target of quinolone antibiotics?

Answer: C

DNA gyrase (a type II topoisomerase) relieves positive supercoils ahead of the replication fork by introducing negative supercoils. Quinolones (e.g., ciprofloxacin) trap the enzyme-DNA complex, blocking replication. [PP: FA2 Sec11 Q1]

Q12. Insertion or deletion of 1–2 nucleotides in a coding sequence causes which type of mutation?

Answer: D

Adding or removing 1 or 2 nucleotides (not multiples of 3) shifts the reading frame for all codons downstream, typically producing a severely altered or truncated protein. [PP: FA2 Sec11 Q6]

Lec 11: Protein Synthesis & Transport Protein

Q13. [PP] Which molecule guides the ribosome-mRNA-nascent protein complex to the endoplasmic reticulum during protein synthesis?

Answer: C

Proteins destined for secretion or membrane insertion carry an N-terminal signal peptide. SRP recognises this signal co-translationally and escorts the entire translation complex to the SRP receptor on the ER membrane. The signal peptide is then cleaved. [PP: SA2 MED35 Q18]

Q14. [PP] In a promoter mutation, what is the PRIMARY (most directly) altered event?

Answer: E

The promoter is the DNA landing site for RNA polymerase and transcription factors. A promoter mutation directly changes BINDING AFFINITY at that site; downstream effects (altered mRNA and protein production) are secondary consequences. [PP: SA2 MED35 Q43, Copy of 4.pdf Q7]

Q15. [PP] Which antibiotic blocks translation in BOTH prokaryotes AND eukaryotes?

Answer: C

Puromycin is an aminoacyl-tRNA analogue that enters the A site of both 70S and 80S ribosomes, forms a peptide bond with the growing chain, and causes premature chain release. It is used experimentally and occasionally clinically. [PP: Key Simulation Q6, Re-exam Q25]

Q16. [PP] Which protein does NOT require a signal sequence for proper targeting?

Answer: E

Pyruvate kinase is a cytosolic glycolytic enzyme — it remains in the cytoplasm and does not require targeting. Insulin, growth hormone (secreted), GLUT4 (membrane), and Cytochrome P450 (ER) all require signal or targeting sequences. [PP: SA2 Med34 Q7 context, Re-exam Q20]

Q17. [PP] Given mRNA: 5'-AUG UGG CAU CGU UAA-3'. How many amino acids does the translated peptide contain?

Answer: B

AUG (Met) – UGG (Trp) – CAU (His) – CGU (Arg) – UAA (STOP). The stop codon is not translated into an amino acid. Peptide = Met-Trp-His-Arg = 4 amino acids. [PP: FA2 Sec11 Q4, similar in MED35 papers]

Q18. [PP] Tetracycline exerts its antibacterial effect by:

Answer: B

Tetracycline chelates Mg^{2+} and binds the 30S ribosomal subunit near the A site, preventing aminoacyl-tRNA from entering. Elongation is halted. It is bacteriostatic. [PP: Med36 section, Re-exam Q20]

Q19. [PP] Chloramphenicol inhibits bacterial translation by blocking:

Answer: B

Chloramphenicol binds the 50S subunit and inhibits peptidyl transferase (the 23S rRNA catalytic activity that forms peptide bonds). It also inhibits mitochondrial 70S ribosomes, explaining some human toxicity (aplastic anaemia). [PP: Med36, Re-exam Q21]

Q20. [PP] Eukaryotic ribosome size is:

Answer: D

Eukaryotic ribosome: 80S = 40S (small) + 60S (large). Prokaryotic: 70S = 30S + 50S. Mitochondrial ribosomes (~55S) are more similar to prokaryotic ribosomes, explaining why some antibiotics targeting 70S have mitochondrial side effects. [PP: Key Simulation Lec11 Q1]

**Q21. [PP] From the following mRNA sequence, how many amino acids are found in Peptide X?
5'-GGGAUG CCU AAU GAG UGA CCU-3'**

Answer: B

Start at AUG: AUG (Met) – CCU (Pro) – AAU (Asn) – GAG (Glu) – UGA (STOP). Peptide = Met-Pro-Asn = 3 amino acids. The GGG before AUG is the 5'-UTR; translation starts at AUG. [PP: Med35-36 style question]

Q22. [PP] Which elongation factor escorts aminoacyl-tRNA to the ribosome A site during translation elongation?

Answer: C

EF-Tu (prokaryotes)/eEF1A (eukaryotes) forms a ternary complex with aminoacyl-tRNA and GTP, delivering the charged tRNA to the A site. GTP hydrolysis triggers conformational change releasing EF-Tu-GDP. EF-Ts regenerates EF-Tu-GTP. EF-G mediates translocation. [PP: SA2 Med34 Q4, Re-exam Q19]

Q23. In eukaryotic protein translation, the initiator Met-tRNA occupies which ribosomal site during initiation?

Answer: C

The initiator Met-tRNA is unique in directly occupying the P site during assembly of the initiation complex. This is catalysed by eIF2-GTP. All subsequent aminoacyl-tRNAs enter at the A site. [PP: Re-exam Q21, Med36 style]

Q24. Post-translational modifications include all of the following EXCEPT:

Answer: E

Post-translational modifications (PTMs) occur AFTER translation: glycosylation, phosphorylation, ubiquitination, acetylation, signal peptide cleavage, disulfide bond formation, etc. Codon degeneracy (multiple codons for one amino acid) is a property of the genetic code, not a PTM. [PP: Med35-36 style]

Lec 12: Liver Function

Q25. [PP] Which is the rate-limiting enzyme of bile acid synthesis?

Answer: B

CYP7A1 catalyses the first committed step converting cholesterol to bile acids. Found only in hepatocytes. Upregulated by cholesterol; downregulated by bile salts (feedback inhibition). Statins target HMG-CoA reductase (cholesterol synthesis — a separate pathway). [PP: SA2 MED35 Q66, SA2 Med34]

Q26. [PP] Which molecule conjugates bilirubin to make it water-soluble for excretion?

Answer: C

UDP-glucuronyl transferase (UGT1A1) in hepatocytes adds glucuronic acid to unconjugated bilirubin → bilirubin diglucuronide (conjugated, water-soluble) → excreted into bile. Deficiency: Crigler-Najjar (severe) or Gilbert's syndrome (mild). [PP: FA2 Med33-34 Q12, SA2 MED35 Q88]

Q27. [PP] In thalassemia, the predominant form of elevated serum bilirubin is:

Answer: B

Thalassemia causes haemolysis → excess haem → unconjugated bilirubin overwhelms the liver's conjugation capacity. Pre-hepatic (haemolytic) jaundice = elevated unconjugated bilirubin. [PP: FA2 Sec11 Q10]

Q28. [PP] A 50-year-old with jaundice and bile duct gallstones most likely shows elevation of:

Answer: B

Post-hepatic (obstructive) jaundice = bile duct obstruction. GGT and ALP are released from biliary epithelial cells when bile backs up. ALT/AST = hepatocellular injury markers. Normal ALT/AST + elevated ALP/GGT = cholestatic pattern. [PP: SA2 MED35 Q63]

Q29. [PP] Disulfiram (Antabuse) treats alcoholism by inhibiting:

Answer: B

Ethanol → (ADH) → acetaldehyde → (ALDH, mitochondria) → acetate. Disulfiram blocks ALDH → acetaldehyde accumulates → flushing, tachycardia, nausea, vomiting. This aversive reaction discourages drinking. [PP: SA2 MED35, Med36, Re-exam Q19]

Q30. [PP] Excess NADH generated by ethanol metabolism most directly causes:

Answer: C

Ethanol → large NADH excess. High NADH/NAD⁺: inhibits gluconeogenesis, inhibits TCA cycle, inhibits beta-oxidation, promotes fatty acid synthesis and lactic acidosis. Net: triglyceride accumulation → fatty liver → alcoholic hepatitis → cirrhosis. [PP: SA2 Med34 Q8, FA2 Sec11 Q11]

Q31. [PP] Which is the tumour marker for liver cancer (hepatocellular carcinoma)?

Answer: C

AFP is normally produced by fetal liver; very low in adults (<10 ng/mL). Markedly elevated AFP in an adult suggests hepatocellular carcinoma (HCC). Also elevated in testicular/ovarian germ cell tumours and pregnancy. [PP: SA2 Med34 Q6, Re-exam Q10]

Q32. [PP] Which is a secondary bile acid (formed in the intestine by bacterial action)?

Answer: C

Primary bile acids (synthesised in liver): cholic acid and chenodeoxycholic acid. Secondary bile acids (intestinal bacteria remove a 7-OH group): deoxycholic acid (from cholic) and lithocholic acid (from chenodeoxycholic). Glycocholic/taurocholic are conjugated primary acids. [PP: SA2 Med34 Q7, FA2 Sec11 Q12]

Q33. [PP] Phase II biotransformation in the liver is characterised by:

Answer: C

Phase I (CYP450): introduces a reactive group via oxidation/reduction/hydrolysis. Phase II (transferases): conjugation with glucuronic acid, glutathione, sulphate, acetate, or amino acids → increases water-solubility for excretion. Key Phase II enzyme: Glutathione S-Transferase (GST), UGT. [PP: SA2 Med34 Q3, Med36 Q7]

Q34. [PP] Crigler-Najjar syndrome Type I results from complete deficiency of:

Answer: C

Without UGT1A1, bilirubin cannot be conjugated → severe unconjugated hyperbilirubinemia from birth → kernicterus (brain damage) → fatal without liver transplant. Type II = partial deficiency (responds to phenobarbital). Gilbert's = 10–30% activity, usually benign. [PP: Med36 Q6]

Q35. [PP] The enzyme responsible for the MEOS (Microsomal Ethanol Oxidising System) is:

Answer: C

MEOS (in smooth ER) is induced at high ethanol concentrations and with chronic use. The key enzyme is CYP2E1. Uses NADPH and O₂, produces ROS → oxidative stress and liver damage. Explains why chronic drinkers metabolise alcohol faster (CYP2E1 induction). [PP: Med36 Q5]

Q36. Which test BEST reflects hepatic SYNTHETIC function?

Answer: D

Synthetic function = what the liver makes. Albumin (exclusively by hepatocytes) and clotting factors (PT reflects factors II, VII, IX, X). ALT/AST = hepatocellular injury. GGT/ALP = biliary markers. Bilirubin = excretory function.

Q37. Heme oxygenase catalyses which reaction in bilirubin metabolism?

Answer: C

Heme oxygenase (reticuloendothelial system): haem + O₂ + NADPH → biliverdin (green) + CO + Fe²⁺. This ring-opening reaction is rate-limiting for bilirubin production. Biliverdin reductase then converts biliverdin → bilirubin (yellow). 85% of bilirubin comes from senescent RBC haemoglobin.

Q38. In which type of jaundice is stool pale (clay-coloured) and urine dark, with urobilinogen absent from urine?

Answer: C

Post-hepatic (obstructive) jaundice: bile duct blocked → conjugated bilirubin cannot reach intestine → no urobilinogen formed → pale stool. Conjugated bilirubin backs up into blood → excreted in urine (dark urine). Pre-hepatic: increased urobilinogen, dark stool.

Q39. Enterohepatic circulation of bile acids refers to:

Answer: B

~95% of bile acids secreted into the intestine are reabsorbed at the terminal ileum, returned to the liver via the portal vein, re-conjugated, and re-secreted. Only 5% are newly synthesised. This conserves the bile acid pool (recycled 6–10×/day). Cholestyramine binds bile acids → prevents reabsorption → lowers cholesterol.

Q40. AST/ALT ratio >2 is most suggestive of:

Answer: C

AST is found in liver, heart, muscle, and RBCs; ALT is more liver-specific. In alcoholic liver disease/cirrhosis, AST > ALT (ratio >2) because alcohol preferentially damages mitochondria releasing AST, and alcohol also depletes pyridoxal-5-phosphate (cofactor for ALT). In viral hepatitis: ALT > AST (ratio <1). [PP: Copy of 4.pdf Q49]

Lec 13: Thalassemia

Q41. [PP] What is the genotype of Hb Bart's hydrops fetalis?

Answer: C

Hb Bart's = ALL 4 alpha-globin genes absent (α⁰α⁰). Without alpha chains, gamma chains form tetramers (gamma₄ = Hb Bart's) with extremely high O₂ affinity → severe tissue hypoxia → fatal in utero (hydrops fetalis). [PP: SA2 MED35, SA2 Med34, Med36, Re-exam]

Q42. [PP] Which genotype results in Hb H disease?

Answer: B

Hb H disease = 3 of 4 alpha genes non-functional (α⁻α⁻α). Only ONE functional alpha gene remains. Excess beta chains form beta₄ tetramers = Hb H. Causes moderate-to-severe haemolytic anaemia. [PP: SA2 Med34, Key Simulation Q4]

Q43. [PP] Hydroxyurea is used in beta-thalassemia because it:

Answer: B

Hydroxyurea (also 5-azacytidine, butyrate, erythropoietin) reactivates fetal haemoglobin (HbF, α₂γ₂) production. Increased HbF compensates for deficient HbA in beta-thalassemia and sickle cell disease. Deferoxamine/deferiprone are iron chelators. [PP: SA2 Med34, Key Simulation Q2&Q5;]

Q44. [PP] Father has beta-thalassemia trait (carrier); mother has homozygous HbE. What is the probability their child has thalassemia disease?

Answer: C

Father: β/β carrier (50% chance of passing β). Mother: β/β (always passes β). Child who gets β (from father) + β (from mother) = β/β thalassemia disease (50%). Child who gets β (from father) + β = HbE trait (50%). [PP: SA2 Med34 Q2, Med36 Q5]

Q45. [PP] Hemoglobin Constant Spring (HbCS) is caused by which mutation?

Answer: C

HbCS: the normal alpha-globin stop codon (UAA, codon 142) mutates to CAA (Glutamine) → translation continues past the normal stop → abnormally long alpha chain (172 aa instead of 141). Non-deletion alpha-thalassemia. Common in Thailand (1–8%). [PP: SA2 MED35, Key Simulation Q4]

Q46. [PP] Which haemoglobin predominates in FETAL circulation?

Answer: B

HbF (alpha γ) dominates from 2nd trimester to birth (~97% at birth). Its higher O $_2$ affinity than HbA facilitates O $_2$ transfer from maternal to fetal blood. Replaced by HbA by 6–12 months of age. [PP: SA2 Med34, SA2 MED35 Q50/Q80, Med36]

Q47. [PP] Which haemoglobin is found in the EMBRYO (earliest development, yolk sac stage)?

Answer: E

Embryonic haemoglobins (yolk sac, first ~8 weeks): Hb Gower 1 (zeta ϵ), Hb Gower 2 (alpha ϵ), Hb Portland (zeta γ). Zeta = embryonic alpha-like; epsilon = embryonic beta-like chain. [PP: SA2 Med34 Q1, Key Simulation Q6]

Q48. [PP] Repeated blood transfusions in severe thalassemia most commonly leads to:

Answer: B

Each unit of blood contains ~200–250 mg iron. The body cannot excrete excess iron → accumulates in heart (cardiomyopathy), liver (cirrhosis), pancreas (diabetes), pituitary (hypogonadism). Iron chelation therapy (deferixamine/deferiprone/deferasirox) is mandatory. [PP: Med36, Lec13]

Q49. In beta-zero thalassemia/HbE (most common severe thalassemia in Thailand), the HbE mutation at codon 26 (GAG→AAG) also creates:

Answer: B

HbE mutation (codon 26: Glu→Lys) also activates a cryptic splice site in exon 1 → abnormal splicing → reduced normal beta-globin mRNA. Combined with beta 0 -thalassemia = severe disease requiring transfusions. This is the most prevalent severe haemoglobin disorder in Southeast Asia.

Q50. Alpha-thalassemia trait (2 functional alpha genes remain) can result from which genotype(s)?

Answer: C

Alpha-thal trait = 2 of 4 alpha genes deleted. Can be CIS ($-\alpha/\alpha$: both deletions on same chromosome, alpha-thal 1 heterozygote) or TRANS ($-\alpha/-\alpha$: one deletion per chromosome, alpha-thal 2 homozygote). Both give mild/no anaemia, but CIS carriers can have Hb Bart's babies if both parents carry $-\alpha$.

Lec 14: Apoptosis

Q51. [PP] Which is a characteristic of APOPTOSIS but NOT necrosis?

Answer: B

Apoptosis (programmed, energy-dependent): cell shrinkage, chromatin condensation, nuclear fragmentation, membrane blebbing → apoptotic bodies phagocytosed WITHOUT inflammation. Necrosis: cell swelling, membrane rupture, contents released → inflammation. [PP: SA2 Med34, Med36 Q4, Re-exam Q6]

Q52. [PP] Which component is NOT part of the apoptosome?

Answer: E

Apoptosome (intrinsic pathway) = Cytochrome C + APAF-1 + dATP → wheel-like heptamer that recruits and activates pro-caspase 9. Pro-caspase 10 is involved in the EXTRINSIC pathway (DISC complex after death receptor ligation). [PP: SA2 Med34 Q4, Key Simulation Q4, Re-exam Q8]

Q53. [PP] Which MAPK signalling kinase INDUCES apoptosis?

Answer: B

MAPK family: ERK1/2 = survival and proliferation (anti-apoptotic in most contexts). p38 MAPK and JNK = stress-activated by UV, cytokines, oxidative stress → pro-apoptotic. AKT is the major survival kinase (phosphorylates BAD → inactivates it). [PP: SA2 Med34 Q3, Med36 Q5, Re-exam Q3]

Q54. [PP] Which member of the Bcl-2 family is PRO-apoptotic?

Answer: B

Anti-apoptotic: Bcl-2, Bcl-xL, Mcl-1, Bcl-w (suppress death). Pro-apoptotic multidomain: BAX, BAK (form pores in outer mitochondrial membrane → Cytochrome C release). BH3-only: BAD, BID, PUMA, NOXA. [PP: SA2 Med34 Q7, Key Simulation Q7, Re-exam Q2]

Q55. [PP] Hoechst 33342 staining is used in apoptosis research to assess:

Answer: C

Hoechst 33342 binds AT-rich DNA minor groove. Under fluorescence microscopy, apoptotic cells show condensed, fragmented, brightly stained nuclei vs. diffuse normal staining. Annexin V detects PS externalisation; PI (flow cytometry) measures DNA content. [PP: SA2 Med34 Q5, SA2 MED35, Re-exam Q9]

Q56. [PP] In flow cytometric cell cycle analysis, apoptotic cells appear as which population?

Answer: B

Apoptotic DNA is cleaved into nucleosome-sized fragments. Small fragments wash out during staining → less DNA than G1 cells → 'sub-G1' population to the LEFT of the G1 peak in a DNA content histogram. Quantitative measure of apoptosis by flow cytometry. [PP: SA2 Med34 Q6, Re-exam Q10]

Q57. [PP] The KEY initiator caspase of the EXTRINSIC (death receptor) apoptosis pathway is:

Answer: D

Extrinsic pathway: death receptor ligation (Fas-FasL, TRAIL, TNF) → DISC formation → pro-caspase 8 is recruited and activated. Active caspase 8 then cleaves executioner caspases (3, 7) or BID (linking to intrinsic pathway). Caspase 9 = intrinsic pathway initiator. [PP: SA2 Med34, Key Simulation Q3]

Q58. [PP] Which technique detects apoptosis by identifying phosphatidylserine externalisation?

Answer: C

In early apoptosis, phosphatidylserine (PS) flips from the inner to outer leaflet of the plasma membrane. Annexin V is a Ca^{2+} -dependent protein with high affinity for PS. Fluorescent Annexin V + PI flow cytometry: Annexin+/PI- = early apoptosis; Annexin+/PI+ = late apoptosis/necrosis. [PP: Lec14, Re-exam]

Q59. PARP cleavage by caspases during apoptosis results in:

Answer: B

Active PARP detects and repairs DNA strand breaks (cell survival). Caspases 3 and 7 CLEAVE PARP (116 kDa) → 89 kDa + 27 kDa fragments → INACTIVATES DNA repair, committing the cell to apoptosis. PARP cleavage is a hallmark of caspase activation detectable by Western blot. [PP: Key Simulation, Lec14]

Q60. Anti-apoptotic Bcl-2 family proteins (e.g., Bcl-2, Bcl-xL) prevent apoptosis by:

Answer: B

Bcl-2/Bcl-xL/Mcl-1 BIND and sequester pro-apoptotic BAX and BAK, preventing them from oligomerising and forming pores in the outer mitochondrial membrane. Without pores, cytochrome C stays in the intermembrane space → no apoptosome → no caspase 9 activation → no apoptosis.

Lec 15: Human Genome & Gene Regulation

Q61. [PP] Fragile X syndrome CGG repeat expansion in FMR1 primarily disrupts which process?

Answer: D

When CGG repeats >200, the FMR1 promoter becomes hypermethylated → heterochromatin → transcription SILENCED. No FMRP protein → intellectual disability, autism-like features, macro-orchidism. [PP: FA2 Med33-34 Q7, Re-exam Q1]

Q62. [PP] A 25-year-old female with infertility, no menarche, webbed neck, and broad chest. Most likely karyotype?

Answer: B

These features describe Turner syndrome (monosomy X). Karyotype: 45,XO. Note: 46,XO does not exist; 46 with one X is written as 46,XX,del(Xp). Normal intelligence, webbed neck, broad chest, no menstruation, infertility. [PP: FA2 Sec11 Q9, Key Simulation Q1]

Q63. [PP] The Mediator complex in eukaryotic transcription functions as:

Answer: B

Mediator is a multi-subunit complex that physically bridges activator proteins (bound to distal enhancers) with general transcription factors (at core promoter/TATA box), facilitating RNA pol II recruitment and transcription initiation. [PP: Key Simulation Q5]

Q64. [PP] Estrogen receptor (ER) is classified as which receptor type?

Answer: A

Estrogen (lipid-soluble) crosses the plasma membrane → binds cytoplasmic ER → ER dimerises and translocates to nucleus → binds estrogen response elements (EREs) in DNA → activates gene transcription. ER is a nuclear receptor superfamily member. [PP: SA2 MED35 Q12]

Q65. [PP] A promoter mutation most DIRECTLY alters which event?

Answer: E

The promoter is the DNA sequence where RNA polymerase and transcription factors assemble. A mutation directly changes the binding affinity of these proteins at this site — the first and most direct effect. Altered mRNA (C) and protein (D) are downstream consequences. [PP: SA2 MED35 Q43, Copy of 4.pdf Q7]

Q66. [PP] For MAXIMAL lac operon expression in E. coli, which conditions are required?

Answer: B

Lac operon requires TWO simultaneous conditions: (1) Lactose present → allolactose inactivates repressor (operator is free); (2) Glucose absent → high cAMP → cAMP-CAP complex binds CAP site → strongly activates transcription. [PP: Key Simulation Q6]

Q67. [PP] Klinefelter syndrome (47,XXY) is caused by:

Answer: C

Klinefelter (47,XXY): non-disjunction during maternal or paternal meiosis I or II → gamete with extra sex chromosome → 47,XXY offspring. Features: tall, small testes, azoospermia, gynecomastia, normal-to-mild cognitive effects. [PP: Lec15, SA2 Med36]

Q68. [PP] A riboswitch controls gene expression at the TRANSCRIPTIONAL level when metabolite binding causes:

Answer: B

Transcriptional riboswitch (e.g., TPP riboswitch): metabolite binds aptamer domain → expression platform forms Rho-independent terminator stem-loop (+ run of U residues) → RNA polymerase terminates prematurely → gene OFF. Without metabolite: antiterminator forms → gene ON. [PP: SA2 Med34 Q5]

Q69. [PP] AGO2-mediated mRNA cleavage in the siRNA pathway requires:

Answer: A

siRNA pathway (AGO2, unique slicer activity): PERFECT complementarity between siRNA and mRNA → AGO2 cleaves target mRNA at the centre of the duplex. miRNA pathway (AGO1–4): IMPERFECT complementarity → translational repression or deadenylation. [PP: FA2 Sec11 Q8]

Q70. DNA methylation at CpG promoter regions typically results in:

Answer: C

CpG methylation by DNMTs → recruits methyl-CpG binding proteins → recruit HDACs → histone deacetylation → condensed chromatin (heterochromatin) → gene SILENCING. TET enzymes demethylate DNA → gene reactivation. This is epigenetic regulation.

Q71. Which feature makes VNTR/STR markers useful for DNA fingerprinting?

Answer: C

VNTRs and STRs are polymorphic — the NUMBER of tandem repeats varies between individuals. Analysing multiple loci simultaneously creates a unique 'fingerprint' pattern. Used in forensics, paternity testing, and the CODIS database. [PP: Lec19, Re-exam Q27]

Q72. Chromosomal notation for Down syndrome (trisomy 21) in a female is:

Answer: C

Down syndrome: 47 chromosomes with an extra chromosome 21. Female: 47,XX,+21. Male: 47,XY,+21. Most common form: non-disjunction of chr 21 during meiosis (94%). Translocation Down syndrome: 46 total chromosomes but extra chr 21 material fused to chr 14 [46,XX,t(14;21)].

Lec 16: Blood & Heme Metabolism

Q73. [PP] The rate-limiting enzyme of haem synthesis is:

Answer: B

ALAS catalyses the first committed step: glycine + succinyl-CoA → delta-ALA. Regulated by haem via feedback inhibition. Requires pyridoxal phosphate (PLP, Vitamin B6) as cofactor. Located in the mitochondrial matrix. [PP: SA2 Med34 Q2, Re-exam Q6]

Q74. [PP] Which TCA cycle compound is the direct precursor for haem synthesis?

Answer: C

ALAS condenses succinyl-CoA (from TCA cycle) + glycine → ALA. This links energy metabolism (TCA cycle) to haem biosynthesis. [PP: FA2 Sec11 Q13, Re-exam Q11]

Q75. [PP] Lead (Pb) poisoning inhibits which TWO enzymes in haem synthesis?

Answer: B

Lead inhibits: (1) ALAD (2 ALA → uroporphobilinogen) → ALA accumulates in urine (neurotoxic); (2) Ferrochelatase (Fe²⁺ insertion into protoporphyrin IX → haem) → zinc protoporphyrin accumulates. Result: microcytic anaemia with basophilic stippling. [PP: SA2 Med34 Q3, BCH212 Re-exam Q65]

Q76. [PP] Porphyrria Cutanea Tarda (PCT), the most common porphyria, is caused by deficiency of:

Answer: D

PCT: UROD deficiency → uroporphyrinogens accumulate → PHOTSENSITIVITY (blistering, scarring in sun-exposed skin), hyperpigmentation. NO neurological symptoms (unlike AIP). Triggers: alcohol, HCV, iron excess, oestrogens. Most common porphyria. [PP: FA2 Sec11 Q14, Re-exam Q1]

Q77. [PP] Acute Intermittent Porphyria (AIP) is characterised by all of the following EXCEPT:

Answer: D

AIP (porphobilinogen deaminase deficiency): ALA and PBG accumulate → NEUROLOGICAL symptoms — severe abdominal pain, peripheral neuropathy, psychiatric symptoms, port-wine urine from PBG oxidation. NO photosensitivity (porphyrins don't accumulate in skin). Barbiturates induce ALAS → worsen attacks. [PP: Key Simulation, Re-exam Q8]

Q78. [PP] Which enzyme in haem synthesis requires Vitamin B6 (pyridoxal phosphate) as cofactor?

Answer: A

ALAS (glycine + succinyl-CoA → ALA) is a PLP-dependent enzyme — forms a Schiff base with glycine. B6 deficiency → impaired haem synthesis → sideroblastic anaemia. [PP: FA2 Sec11 Q15, BCH212 Re-exam Q4]

Q79. [PP] Mature red blood cells CANNOT synthesise haem because they:

Answer: B

Mature RBCs have neither nucleus nor mitochondria. ALAS (the first step) is located in the MITOCHONDRIAL MATRIX. Without mitochondria, haem synthesis cannot begin. Haem is made during erythroblast development in the bone marrow. [PP: SA2 Med34 Q1, Re-exam Q22]

Q80. Which amino acid, together with succinyl-CoA, is the starting material for haem synthesis?

Answer: C

ALAS condenses glycine (provides C1 and the nitrogen of ALA) + succinyl-CoA (4 carbons) → ALA (delta-aminolevulinic acid, 5 carbons). Two ALA molecules then condense to form porphobilinogen (catalysed by ALAD).

Lec 17: Plasma Protein & Blood Clotting Factor

Q81. [PP] The MAJOR plasma protein and its primary function are:

Answer: B

Albumin: ~60% of total plasma protein (3.5–5 g/dL), synthesised exclusively by the liver (12 g/day). Primary function: maintains colloid osmotic pressure (keeps fluid in vessels). Also transports fatty acids, bilirubin, hormones, and drugs. Half-life: 15–20 days. [PP: FA2 Sec11 Q18]

Q82. [PP] The FIRST step in haemostasis after blood vessel injury is:

Answer: C

Haemostasis sequence: (1) Vasoconstriction — local reflex reduces blood flow; (2) Primary haemostasis — platelet adhesion (vWF-GPIb/IX-collagen) + aggregation (GPIIb/IIIa-fibrinogen) = platelet plug; (3) Secondary haemostasis — coagulation cascade → fibrin; (4) Fibrinolysis. [PP: FA2 Sec11 Q16]

Q83. [PP] Which enzyme digests fibrin during fibrinolysis?

Answer: B

Plasmin (serine protease) cleaves fibrin → fibrin degradation products. Plasminogen → plasmin is catalysed by t-PA (tissue plasminogen activator) and urokinase. Thrombin converts fibrinogen → fibrin (opposite). [PP: FA2 Sec11 Q17, Med36]

Q84. [PP] Alpha-1 antitrypsin (AAT) deficiency primarily causes:

Answer: C

AAT inhibits neutrophil elastase in the lungs. Without AAT, elastase degrades lung elastin → emphysema (panacinar, lower lobes). Misfolded Z-type AAT also accumulates in hepatocytes → cirrhosis. Smoking inhibits AAT-elastase binding → worsens disease. [PP: Med36 Q3, Lec16–17]

Q85. [PP] Multiple myeloma is associated with elevated levels of which plasma protein fraction?

Answer: D

Multiple myeloma = malignant plasma cell proliferation producing monoclonal immunoglobulin (M-protein). Serum electrophoresis: narrow M-spike in gamma region. Gamma globulins = IgG, IgA, IgM, IgD, IgE — produced by plasma cells (B lymphocytes). [PP: Med36 Q1]

Q86. [PP] Aspirin prevents arterial thrombosis by:

Answer: B

Aspirin irreversibly acetylates COX-1 (and COX-2) → prevents TXA2 synthesis in platelets. TXA2 normally promotes platelet aggregation and vasoconstriction. Platelets have no nucleus → cannot resynthesise COX → effect lasts platelet lifetime (~7–10 days). [PP: SA2 MED35 Q11, Key Simulation Q2]

Q87. [PP] Vitamin K deficiency most directly leads to:

Answer: B

Vitamin K is a cofactor for gamma-glutamyl carboxylase, which gamma-carboxylates factors II (prothrombin), VII, IX, X (and proteins C, S). Without carboxylation, these factors cannot bind Ca^{2+} → non-functional → prolonged PT. Warfarin/dicoumarol inhibit vitamin K recycling. [PP: SA2 MED35 Q45, Re-exam Q23]

Q88. [PP] Haemophilia A is caused by deficiency of:

Answer: C

Haemophilia A (most common, ~80% of haemophilia): X-linked recessive, Factor VIII deficiency. Prolonged aPTT (intrinsic pathway defect). Haemophilia B (Christmas disease): Factor IX deficiency, also X-linked. Both present with haemarthrosis and easy bruising. [PP: Lec16–17]

Q89. [PP] Von Willebrand factor (vWF) plays what primary role in haemostasis?

Answer: B

vWF (multimeric glycoprotein from endothelium and platelets) bridges platelets (via GP Ib/IX) to exposed collagen at the vessel injury site. Also carries and protects Factor VIII from premature degradation. Von Willebrand disease = most common inherited bleeding disorder. [PP: Lec16–17]

Q90. Haptoglobin's main function in plasma is to:

Answer: B

Haptoglobin (alpha₂-globulin) binds free Hb from lysed RBCs → Hb-Hp complex cleared by the reticuloendothelial system, preventing iron loss in urine and kidney damage from free Hb. In haemolysis: haptoglobin is consumed → low serum haptoglobin = marker of haemolysis.

Q91. Albumin does NOT transport which of the following?

Answer: C

Albumin transports: fatty acids, unconjugated bilirubin, drugs (warfarin, phenytoin, NSAIDs), steroid hormones, calcium. IRON transport is primarily by TRANSFERRIN (a beta-globulin). Not albumin. [PP: BCH212 Re-exam Q37]

Q92. t-PA (tissue plasminogen activator) causes:

Answer: C

t-PA is produced by endothelial cells and cleaves plasminogen → plasmin, which dissolves fibrin clots (fibrinolysis). t-PA is fibrin-bound → locally active at clot sites. Recombinant t-PA (alteplase, tenecteplase) = thrombolytic therapy for acute MI and ischaemic stroke. [PP: Med36 Q7]

Lec 18: Nutrition

Q93. [PP] Pellagra (dermatitis, diarrhoea, dementia) is caused by deficiency of:

Answer: C

Pellagra = Niacin (B3) deficiency. The 4 Ds: Dermatitis (photosensitive symmetrical rash), Diarrhoea, Dementia, Death. Niacin forms NAD⁺/NADP⁺. Can be synthesised from tryptophan (60 mg Trp → 1 mg niacin), requiring B6. Isoniazid → B6 deficiency → secondary pellagra. [PP: SA2 Med34 Q1, Re-exam Q36]

Q94. [PP] Prolonged consumption of raw egg whites causes deficiency of:

Answer: D

Raw egg whites contain AVIDIN, a glycoprotein that binds biotin (B7) with very high affinity, blocking gut absorption. Cooking denatures avidin. Biotin is a cofactor for carboxylase enzymes (pyruvate carboxylase, acetyl-CoA carboxylase). Deficiency: alopecia, dermatitis, neurological symptoms. [PP: SA2 Med36 Q4, FA2 Med33-34 Q23, Re-exam Q5]

Q95. [PP] Macrocytic anaemia + elevated homocysteine + NORMAL methylmalonic acid indicates:

Answer: B

Both B12 and folate deficiency cause macrocytic anaemia + elevated homocysteine. KEY differentiator: methylmalonic acid (MMA). B12 deficiency → MMA ELEVATED (methylmalonyl-CoA mutase needs B12). Folate deficiency → MMA NORMAL. B12 deficiency also causes neurological symptoms (subacute combined degeneration). [PP: SA2 Med36 Q3]

Q96. [PP] Vitamin D deficiency results in:

Answer: B

Vitamin D → increases intestinal Ca²⁺ and phosphate absorption + bone mineralisation. Deficiency → low Ca²⁺ + low PO₄ → secondary hyperparathyroidism → impaired bone mineralisation → Rickets (children, bowed legs) or Osteomalacia (adults, bone pain). Hypercalcaemia = Vitamin D TOXICITY. [PP: SA2 Med34 Lec18 Q3]

Q97. [PP] Vitamin C is required as cofactor for which enzyme in collagen synthesis?

Answer: B

Prolyl hydroxylase and lysyl hydroxylase require Vitamin C to maintain iron in the active Fe²⁺ state. These enzymes hydroxylate proline and lysine residues in collagen → essential for triple helix stability and crosslinking. Deficiency → SCURVY: bleeding gums, perifollicular haemorrhages, poor wound healing. [PP: SA2 Med34 Q2, Re-exam Q12]

Q98. [PP] Cretinism in newborns results from severe maternal deficiency of:

Answer: B

Iodine → thyroid hormone (T3, T4) synthesis. Severe iodine deficiency during pregnancy → maternal and fetal hypothyroidism → cretinism: severe intellectual disability, stunted growth, deaf-mutism, spastic diplegia. Common in mountainous/inland areas far from sea. [PP: SA2 MED35 Q71, SA2 Med34 Lec21 Q7, Re-exam Q2]

Q99. [PP] Neural tube defects (spina bifida, anencephaly) result from deficiency of:

Answer: A

Folate is essential for neural tube closure (days 21–28 after conception — often before pregnancy is known). Deficiency → spina bifida, anencephaly, encephalocele. Recommended: 400–800 µg/day for all women of childbearing age. [PP: SA2 MED35, Re-exam Q10, Copy Q7]

Q100. [PP] Isoniazid (INH) treatment for TB most directly causes secondary deficiency of:

Answer: C

Isoniazid inhibits pyridoxal kinase → cannot activate Vitamin B6 to PLP → B6 deficiency → peripheral neuropathy, sideroblastic anaemia, secondary pellagra (PLP needed for niacin synthesis from tryptophan). Prevention: give pyridoxine (B6) supplements with INH. [PP: BCH212 Re-exam Q15, Copy Q23]

Q101. [PP] The ACTIVE form of Vitamin D produced in the KIDNEY is:

Answer: D

Vitamin D pathway: Skin (UV) → D3 → Liver (25-hydroxylase) → 25-OH-D₃ (calcidiol, storage form, LONG half-life 2–3 weeks, measured clinically) → Kidney (1 α -hydroxylase) → 1,25-(OH)₂-D₃ (calcitriol, ACTIVE form, short half-life 4–6 hours). [PP: SA2 Med34 Lec21 Q9, Copy Q82]

Q102. [PP] Kwashiorkor features oedema but NOT Marasmus because kwashiorkor involves:

Answer: B

Kwashiorkor = primarily protein deficiency (often adequate calories): low albumin → low oncotic pressure → fluid leaks to interstitium → pitting oedema, ascites. Also: moon face, skin lesions, fatty liver, apathy. Marasmus = total calorie restriction: severe wasting/emaciation, NO oedema (albumin often relatively preserved). [PP: SA2 Med34 Lec21 Q8, Med36 Q4, Re-exam Q3]

Q103. [PP] Prolonged PPI (proton pump inhibitor) use most commonly leads to deficiency of:

Answer: C

B12 requires gastric acid to release it from food proteins → binds intrinsic factor (from parietal cells) → absorbed in terminal ileum. PPIs suppress acid → B12 remains bound to food protein → cannot bind IF → B12 malabsorption. Risk increases with >2 years of use. [PP: SA2 Med36 Q1, SA2 Med34 Lec21 Q4]

Q104. [PP] Which statement about fat-soluble vitamins (A, D, E, K) is TRUE?

Answer: B

Fat-soluble vitamins: stored in adipose tissue and liver, NOT easily excreted in urine → can accumulate → hypervitaminosis. Most dangerous: Vitamin A (teratogen in pregnancy, liver damage) and Vitamin D (hypercalcaemia). Water-soluble vitamins (B-complex, C) act as coenzymes and are readily excreted. [PP: FA2 Sec11 Q19]

Q105. Vitamin K is required for gamma-carboxylation of glutamate residues in which clotting factors?

Answer: B

Vitamin K cofactor for gamma-glutamyl carboxylase → adds carboxyl groups to glutamate residues in factors II, VII, IX, X (and anticoagulant proteins C, S). Gamma-carboxyglutamate (Gla) residues bind Ca²⁺ → enables factor-membrane interactions essential for coagulation. Warfarin blocks this cycle.

Q106. Vitamin C acts as an antioxidant primarily by:

Answer: A

Vitamin C (water-soluble) donates electrons to free radicals in the aqueous phase (scavenging) and regenerates active Vitamin E from the tocopheroxyl radical. Vitamin C is then regenerated by glutathione. This forms a cascading antioxidant network: GSH → Vit C → Vit E → membrane lipids.

Q107. A 4-year-old with bleeding gums, perifollicular haemorrhages, poor wound healing, and bone pain. Which vitamin is deficient?

Answer: C

SCURVY = Vitamin C deficiency. All symptoms arise from impaired collagen synthesis (need VitC for prolyl/lysyl hydroxylase): bleeding gums, perifollicular haemorrhages (corkscrew hairs), poor wound healing, periosteal haemorrhages (bone pain). [PP: FA2 Med33-34 Q24]

Q108. Which vitamin is directly involved in converting tryptophan to niacin?

Answer: C

The pathway Tryptophan → kynurenine → 3-hydroxykynurenine → 3-hydroxyanthranilic acid → quinolinic acid → niacin requires PLP (active Vitamin B6) as a cofactor for kynureninase and aminotransferase steps. B6 deficiency → impaired niacin synthesis from Trp → secondary pellagra. [PP: SA2 Med36 Q2, Lec18]

Lec 19: Genetic Engineering

Q109. [PP] Which technique BEST quantifies SARS-CoV-2 viral load?

Answer: D

SARS-CoV-2 has an RNA genome. RT-qPCR: (1) reverse transcriptase converts RNA → cDNA; (2) qPCR amplifies and QUANTIFIES in real time. Ct value inversely correlates with viral load. RT-PCR detects presence/absence but does not quantify. ATK detects antigen (qualitative). [PP: FA2 Sec11 Q22, Med36 Q1]

Q110. [PP] In forensic DNA analysis, which technique creates a DNA 'fingerprint' using VNTR probes?

Answer: B

Southern blot: DNA digested with restriction enzymes → gel electrophoresis → transferred to membrane → probed with labelled VNTR probes → unique banding pattern per individual. Modern forensics uses PCR-based STR analysis (CODIS database). [PP: SA2 MED35 Q29, SA2 Med34 Lec19 Q2, Re-exam Q27]

Q111. [PP] High-throughput gene expression profiling across thousands of genes simultaneously uses:

Answer: D

DNA microarray: mRNA extracted → reverse transcribed → labelled with fluorescent dye → hybridised to array → fluorescence at each spot indicates expression level. Used for cancer gene expression signatures, drug response profiling. [PP: FA2 Sec11 Q23]

Q112. [PP] Which method introduces foreign DNA into cells using an electrical field to create temporary membrane pores?

Answer: B

Electroporation: brief high-voltage electric pulse creates temporary pores in cell membranes. Negatively charged DNA moves through pores (toward anode). Pores reseal after pulse. Used for cells resistant to other transformation methods. [PP: SA2 Med36 Q4, Re-exam Q14]

Q113. [PP] Why must RT-PCR be used to clone a eukaryotic gene for expression in bacteria?

Answer: B

Eukaryotic genes contain introns. Bacteria CANNOT splice mRNA. RT-PCR converts mRNA (already spliced → exons only) → cDNA → PCR amplification → intron-free gene → can be expressed in bacteria to produce the correct protein. [PP: Lec19, Copy Q39, Re-exam Q15]

Q114. [PP] DNA ligase joins two DNA fragments by forming:

Answer: B

DNA ligase seals nicks in the sugar-phosphate backbone by forming a phosphodiester bond between the 3'-OH of one nucleotide and the 5'-phosphate of the adjacent nucleotide, using NAD⁺ (E. coli ligase) or ATP (T4 ligase) as energy. [PP: Key Simulation Lec19 Q2, Re-exam Q30]

Q115. [PP] The FASTEST method to screen bacterial colonies for correct gene insertion is:

Answer: B

Colony PCR: directly pick cells from a colony → add to PCR mix with insert-specific primers → run PCR → positive band = correct clone. Takes ~2–3 hours vs. days for extraction-based methods. [PP: Key Simulation Lec19 Q4, Re-exam Q32]

Q116. [PP] Western blotting detects specific proteins using:

Answer: C

Western blot: proteins separated by SDS-PAGE → transferred to PVDF membrane → blocked → primary antibody (protein-specific) → secondary antibody (enzyme/fluorochrome-conjugated) → signal detection. Detects protein size and expression level. Northern: RNA; Southern: DNA. [PP: Key Simulation Lec19 Q5, Re-exam Q33]

Q117. [PP] Huntington's disease is caused by which mutation in the HTT gene?

Answer: C

Huntington's disease: CAG repeat expansion (>36, normal <36) in HTT exon 1 encodes a polyglutamine (polyQ) tract in huntingtin protein → toxic gain-of-function → neurodegeneration (caudate, putamen). Autosomal dominant with anticipation (repeats expand in successive generations). [PP: SA2 MED35 on Huntington's, Lec19]

Q118. EcoRI cuts 5'-G|AATTC-3'. What type of DNA ends are produced?

Answer: B

EcoRI cuts: 5'-G|AATTC-3' and 3'-CTTAA|G-5' → produces 5' overhangs (-AATT-). These are 5' STICKY ENDS that can anneal with complementary sticky ends from other EcoRI-cut fragments, facilitating directional ligation. Blunt-end cutters: SmaI, EcoRV.

Q119. Which PCR primer design rule is INCORRECT?

Answer: D

Primers should NOT form hairpins or self-complementary structures — these prevent the primer from annealing to the template. Correct rules: ~20 bp; 45–55% GC; G or C at 3' end (GC clamp); matching T_m (within 1°C); avoid repetitive regions and primer-dimer formation. [PP: Key Simulation Lec19 Q, Lec19]

Q120. In blue-white screening, white bacterial colonies indicate:

Answer: B

Blue-white selection: cloning site is WITHIN the lacZ gene. Intact lacZ → beta-galactosidase → cleaves X-gal → BLUE colonies. Gene insert disrupts lacZ → no functional enzyme → WHITE colonies = successful insertion. All colonies must also be antibiotic-resistant (confirms plasmid uptake). [PP: Lec19, Re-exam Q13]

Lec 20: Free Radical & Antioxidant

Q121. [PP] Which enzyme provides the FIRST LINE of enzymatic defence against superoxide (O₂^{•-})?

Answer: C

SOD converts: 2O₂^{•-} + 2H⁺ → H₂O₂ + O₂. Three isoforms: Cu/Zn-SOD (cytoplasm), Mn-SOD (mitochondria), extracellular SOD. H₂O₂ is then detoxified by catalase (peroxisomes) or GPx. SOD is the primary enzymatic defence against superoxide. [PP: SA2 Med34 Lec20 Q1, FA2 Sec11 Q27, Re-exam Q3]

Q122. [PP] Which mineral is incorporated into glutathione peroxidase (GPx)?

Answer: C

GPx contains SELENOCYSTEINE (selenium in place of sulfur) at its active site. GPx uses glutathione (GSH) to reduce H₂O₂ and lipid hydroperoxides. Selenium deficiency → impaired GPx → increased oxidative damage. Mn: in Mn-SOD; Cu+Zn: in Cu/Zn-SOD; Fe: in catalase. [PP: FA2 Sec11 Q26, FA2 Med33-34 Q20, Re-exam Q2]

Q123. [PP] Which vitamin regenerates oxidised Vitamin E (tocopheroxyl radical) back to its active form?

Answer: B

Vitamin E neutralises a lipid peroxyl radical → tocopheroxyl radical (inactive). Vitamin C (water-soluble, in the aqueous phase) donates an electron to regenerate active Vitamin E. Vitamin C is then regenerated by glutathione. Network: GSH → Vit C → Vit E → membrane lipids. [PP: SA2 Med34 Lec20 Q3, Re-exam Q4]

Q124. [PP] A characteristic property of free radicals is all of the following EXCEPT:

Answer: D

Free radicals: extreme reactivity (A), small and diffusible (B), very SHORT lifespan — microseconds to fractions of a second (C), propagate chain reactions (E), damage tissues. Option D is INCORRECT — their lifespan is extremely SHORT, not long. [PP: SA2 Med34 Lec20 Q2, Re-exam Q5]

Q125. [PP] Which part of amino acids is MOST susceptible to attack by free radicals?

Answer: E

Sulfhydryl groups (–SH) of cysteine are the most reactive ROS targets → oxidation to disulfide bonds (S–S), mixed disulfide with glutathione (RS–SG), or sulphenic acid → alters protein structure and function. Aromatic residues (Tyr, Trp) are also vulnerable. [PP: FA2 Sec11 Q25, FA2 Med33-34 Q19, Re-exam Q1]

Q126. Cell MEMBRANE antioxidants include:

Answer: C

Classification by location: PLASMA: Vitamin C, uric acid, bilirubin, beta-carotene, albumin, ceruloplasmin, transferrin. CELL MEMBRANE: alpha-tocopherol (Vit E) and Coenzyme Q10 — both fat-soluble, embedded in membranes, protecting polyunsaturated fatty acids. INTRACELLULAR: SOD, catalase, GPx, glutathione. [PP: Lec20]

Q127. Oxidative stress is defined as:

Answer: C

Oxidative stress: an imbalance where ROS/free radical PRODUCTION exceeds the cell's antioxidant capacity. Normal: ROS production (metabolism, immune function) is balanced by antioxidants. Chronic oxidative stress → cellular damage → contributes to ageing, cancer, cardiovascular disease, neurodegeneration. [PP: Lec20]

Q128. Superoxide anion ($O_2^{\bullet-}$) is PRIMARILY produced by which cellular compartment?

Answer: B

Under normal conditions, ~3–5% of O_2 consumed by mitochondria is incompletely reduced to superoxide at Complex I and Complex III of the ETC (electron leakage). Mitochondria are the MAJOR endogenous source of ROS. Also produced by NADPH oxidase (phagocytes, respiratory burst), xanthine oxidase, CYP450. [PP: Lec20]

Lec 21: Clinical Correlation

Q129. [PP] Which oil has the HIGHEST proportion of saturated fatty acids?

Answer: B

Coconut oil is ~87–92% saturated fatty acids (mainly lauric C12, myristic C14, caprylic C8). Olive oil: ~75% monounsaturated. Canola/soybean/corn: rich in polyunsaturated fatty acids. High saturated fat → raises LDL → cardiovascular risk. [PP: SA2 Med34 Lec21 Q1, Re-exam Q1]

Q130. [PP] The antidote for paracetamol (acetaminophen) overdose is:

Answer: E

Paracetamol overdose: CYP2E1 produces toxic NAPQI → depletes hepatic glutathione → hepatocyte necrosis. NAC: (1) replenishes glutathione (cysteine donor); (2) directly inactivates NAPQI. Best within 8–10 hours. Naloxone: opioid antidote. Fomepizole: methanol/ethylene glycol antidote. [PP: SA2 Med36 Q2, SA2 Med34 Lec21 Q5, Re-exam Q8]

Q131. [PP] After phototherapy for neonatal jaundice, which compound is secreted in BOTH bile and urine?

Answer: D

Phototherapy (blue light, 425–475 nm) converts unconjugated bilirubin → lumirubin (a structural photoisomer, water-soluble WITHOUT needing conjugation by UGT1A1). Lumirubin is excreted in both bile and urine, bypassing the immature hepatic conjugation machinery in neonates. [PP: SA2 Med36 Q1, SA2 Med34 Lec21 Q3, Re-exam Q9]

Q132. [PP] Band keratopathy (calcium deposits in cornea) is associated with which metabolic condition?

Answer: D

Band keratopathy = calcium-phosphate crystal deposits in Bowman's layer of the cornea appearing as a white/grey band. Causes: HYPERCALCAEMIA (hyperparathyroidism, Vitamin D toxicity, sarcoidosis, CKD), chronic ocular inflammation (uveitis). [PP: SA2 Med36 Q3, Lec21 clinical case 11]

Q133. [PP] Which endocrine-disrupting chemical (EDC) is commonly found in polycarbonate plastics?

Answer: C

BPA is a plasticiser in polycarbonate plastics and epoxy resin can linings. It leaches into food/drinks and mimics oestrogen (xenoestrogen) → endocrine disruption. Banned in children's products in many countries. Phthalates are also plastic-derived EDCs. [PP: SA2 Med34 Lec21 Q6, Re-exam Q4]

Q134. [PP] Which form of Vitamin D has the LONGEST half-life in blood?

Answer: B

25-OH-D $\blacksquare$ (calcidiol) half-life: 2–3 weeks → storage form, measured clinically to assess vitamin D status. Active 1,25-(OH) $\blacksquare$ -D $\blacksquare$ (calcitriol) half-life: 4–6 hours (rapidly metabolised). D3/D2 dietary forms: rapidly converted by liver. [PP: SA2 Med34 Lec21 Q9, Copy Q82]

Q135. [PP] Neural tube defects (spina bifida) are caused by deficiency of which nutrient in early pregnancy?

Answer: C

Folate is essential for neural tube closure (days 21–28 after conception). Deficiency → spina bifida (myelomeningocele), anencephaly, or encephalocele. Recommended: 400–800 μ g/day for all women of childbearing age. [PP: SA2 MED35, Re-exam Q10, Copy Q7]

Q136. [PP] Bitot's spots on the conjunctiva represent an early sign of deficiency of:

Answer: B

Vitamin A deficiency progression: Night blindness (earliest, reversible) → Conjunctival xerosis → Bitot's spots (white, foamy triangular keratinised plaques on conjunctiva) → Corneal xerosis → Keratomalacia → Permanent blindness (irreversible). [PP: Lec21 clinical, SA2 MED35 Q35, Copy Q61]

Q137. [PP] In phenylketonuria (PKU), which amino acid is DECREASED in plasma?

Answer: C

PKU: phenylalanine hydroxylase (PAH) deficiency → cannot convert Phe → Tyrosine. Phe ACCUMULATES (neurotoxic). Tyrosine becomes CONDITIONALLY ESSENTIAL → plasma tyrosine DECREASED → reduced dopamine, epinephrine, thyroxine, melanin production. [PP: SA2 MED35 Q16]

Q138. [PP] Standard OGTT (Oral Glucose Tolerance Test) uses what glucose dose?

Answer: C

Standard OGTT: patient fasts 8 hours → drinks 75 g glucose in 250–300 mL water → blood glucose measured at 0, 1, and 2 hours. Diagnostic criteria: DM = 2-hr glucose $\geq$ 200 mg/dL; IGT = 140–199 mg/dL; Normal = <140 mg/dL. [PP: SA2 MED35 Q80]

Q139. A chronic alcoholic presents with oedema, dyspnoea, cardiomegaly, and metabolic acidosis. The most likely vitamin deficiency is:

Answer: B

Wet beriberi = Thiamine (B1) deficiency with cardiovascular features: high-output cardiac failure, cardiomegaly, oedema, tachycardia, lactic acidosis. Alcoholics: poor diet + impaired B1 absorption. Thiamine → TPP (cofactor for pyruvate dehydrogenase, alpha-KG DH, transketolase). Without TPP: pyruvate accumulates → lactic acidosis + impaired cardiac ATP synthesis. [PP: Lec21 clinical case 3]

Q140. Kayser-Fleischer rings + liver disease + neuropsychiatric symptoms. Diagnosis and deficient/elevated substance?

Answer: B

Wilson's disease (ATP7B mutation, autosomal recessive): copper accumulates in liver, brain, cornea, kidney. Kayser-Fleischer rings = copper deposits in Descemet's membrane. Ceruloplasmin markedly LOW; 24-hr urine copper HIGH. Treatment: penicillamine (copper chelator) or zinc (blocks copper absorption). [PP: Lec21 clinical case 17]

Q141. Acanthosis nigricans (dark velvety skin at neck and axillae) in an obese teenager is most associated with:

Answer: B

Benign acanthosis nigricans in young obese patients: insulin resistance → hyperinsulinaemia → insulin binds IGF-1 receptors on keratinocytes → epidermal hyperplasia + hyperpigmentation. Associated with metabolic syndrome and type 2 DM risk. Malignant AN in older, thin patients: rapid onset → consider visceral malignancy (gastric cancer). [PP: Lec21 clinical case 12]

Q142. A patient with gallstones has elevated ALP and GGT but NORMAL ALT and AST. This pattern indicates:

Answer: C

ALP and GGT are enzymes from biliary epithelial cells — elevated in CHOLESTASIS (bile duct obstruction). ALT/AST are hepatocellular INJURY markers. Normal ALT/AST + elevated ALP/GGT = cholestatic pattern, consistent with bile duct obstruction by gallstones. [PP: BCH212 Re-exam Q82, Copy of 4.pdf Q51]

Good luck on your SA2! Prioritise [PP] questions — they are most likely to appear on the real exam.